

**RCIANS ATTENDEASE: AN ATTENDANCE MONITORING SYSTEM USING
GEOFENCING AND FACIAL RECOGNITION TECHNOLOGIES FOR
THE COLLEGE DEPARTMENT OF ROGATIONIST COLLEGE**

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ABSTRACT

This study aimed to develop an automated attendance monitoring system using geofencing and facial recognition technologies for the college department of the Rogationist College. The system comprised two (2) modules: the Office of the Student Affairs (OSA) module, accessed via web application for the creation and monitoring of students' attendance on an event, manage accounts, and verify students' attendance. The student module, accessed via mobile application, acts as an attendance monitoring tool that records students time in and time out, and ensures student's identity through facial recognition.

The development process followed the waterfall methodology of the Software Development Life Cycle (SDLC), which ensured a systematic and structured approach through six (6) phases: requirements analysis, design, development, testing, deployment, and maintenance. The system was evaluated by 10% of the total college students and one representative from the Office of the Student Affairs. The study utilized the International Organization for Standardization 9126 (ISO 9126) quality model, an internationally recognized framework, as the basis for its evaluation. Comments and suggestions gathered during the evaluation were incorporated to verify the system's readiness for deployment.

Upon final evaluation, the system achieved an overall quality score of 0.92, categorized as excellent, demonstrating that it met all functional requirements and quality standards. The findings confirmed that the automated attendance monitoring system using geofencing and facial recognition successfully met its

intended objectives, offering a reliable and efficient solution for the manual process of attendance at Rogationist College.

APPROVAL SHEET

This thesis entitled “**RCians Attendease: An Attendance Monitoring System Using Geofencing and Facial Recognition Technologies for the College Department of Rogationist College**” prepared and submitted by **Jake Matthew A. Viado, Mary Ann J. Anacay, Juilius Patrick A. Peji, Ela Shaina P. Aguilar** in partial fulfillment of the requirements for the course of **Bachelor of Science in Information Technology**, has been examined and is hereby recommended for approval and acceptance.

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Chapter 1

INTRODUCTION

This chapter presents the background of the study, objectives of the study, scope and limitations, significance of the study, and the definition of terms.

Background of the Study

Attendance during events is still monitored manually in many schools, which often leads to several challenges. Manual attendance recording can be time consuming, prone to human error, and difficult to manage, especially during large events where organizers may struggle to monitor students efficiently and ensure that no signature tampering occurs. In addition, the process of generating and organizing attendance records for each student becomes complicated, as organizers need to review and verify one by one.

Implementing an automated attendance record can help ease the process of attendance recording, reducing the time allotted for the signing of students to be able to time in and time out to an event. According to Salih Alih, et al. (2022), comparing the traditional method to an automation of attendance, it reflects domain achievements, and its performances have contributed better to organizations.

At Rogationist College, the current method of monitoring and tracking student attendance involves the use of paper and pen. This manual approach presents several problems in accurately capturing actual student attendance. Based on the personal communication of the researchers with Sempio (2025), the Coordinator of the Office of Student Affairs (OSA), there are frequent cases where

students sign the attendance sheet, but do not stay at the venue for the whole duration of the event. Some students leave shortly after signing in and return only when it is time to sign out. In addition, there have been observations of signature tampering such as students signing in for their classmates or falsifying their time of coming in and going out. These practices make it difficult for OSA to confirm whether students are truly present for the required time. As a result, the integrity of the attendance records is compromised. The OSA also emphasized the need for a more dependable and secure method of tracking attendance.

To address these challenges, the researchers developed an attendance monitoring system utilizing geofencing and facial recognition technologies. The system records student attendance by verifying their identity and physical presence at a specific location, reducing the risk of students signing in on behalf of others and falsifying records. This system aims to improve accuracy and efficiency, minimize human error in report generation, and uphold the integrity of student attendance.

Objective of the Study

The main objective of the study is to develop an attendance monitoring system that utilizes geofencing and facial recognition technologies for the College Department of Rogationist College.

Specifically, the study aimed to:

1. to design the software of an automated attendance monitoring system consisting of two (2) modules: OSA module that manages event details,

- monitor attendance records and manage user accounts and Student module that automatically tracks the event location and marks the attendance and view event details;
2. to develop a reliable automated attendance system that is functional, efficient, and portable that is easy to use and maintain;
 3. to test and evaluate the performance of the system; and
 4. to provide a user manual and system documentation.

Scope and Limitations

This study aimed to design and developed a web-based and mobile application for the tracking of students in events. The application comprises two (2) interconnected modules: the OSA Module and the Student Module.

The OSA Module is a web-based system through which the OSA can manage student accounts, events, and locations, as well as generate and monitor live or finalized attendance reports. Meanwhile, the Student Module is a mobile application for the students. Upon logging in, this module confirms the identity of the user through facial recognition. It also uses the phone's global positioning system (GPS) for student tracking. Additionally, the module is accessible using an Android smartphone device.

However, this study has limitations. It is important to note that the application requires a stable internet connection to accurately retrieve and process the real-time location of the student. Unstable or limited internet connectivity may affect the accuracy of real-time attendance tracking. In addition, the mobile

application is currently available for Android devices only, as iOS integration was excluded due to the complexity of publishing an application to it.

The development of the project took place at Rogationist College and was carried out from August 2025 - January 2026. The developers integrated React and Next.js for the web frontend application, enhancing performance through client-side rendering, optimized routing, and faster load times. React Native with Expo is used for the development of the cross-platform mobile application, while the backend developed by using Java as the programming language and Spring Boot as the backend framework, MongoDB as the database management system. For the implementation of the facial recognition service, the developers used Python with the Open-Source Computer Vision (OpenCV) library. The system was developed using integrated development environments or IDEs; the backend was built in IntelliJ IDEA, an IDE tailored for Java development, while the web and mobile applications were developed using Visual Studio Code, a lightweight and versatile code editor that supports a wide range of extensions and development tools for JavaScript, TypeScript, and Python workflows.

Significance of the Study

The findings of the study could be beneficial specifically to the following:

College Department

The system helps the college department of Rogationist College by providing a way for the College Dean, Cluster Coordinators, and Organization Advisers to accurately monitor students' attendance at every important college

event, enabling them to identify patterns such as habitual absences or tardiness, thereby allowing them to personally reach out and take appropriate action to address any concerns or issues affecting students' attendance behavior.

Students

The system helps the students record their attendance faster and more conveniently. In addition, a system that accurately records attendance records and maintains its integrity, makes students more accountable and responsible for their presence at events.

Office of the Student Affairs

The system helps the Office of the Student Affairs in streamlining attendance tracking for school events and activities, supervising and addressing attendance concerns.

Organization Advisers

The system helps the Organization Advisers by providing an accurate way to monitor their student's attendance and address any concerns regarding the attendance within their respective organization.

Future Researchers

The system can be used by the other researchers to gain an idea when conducting research related to the system.

Definition of Terms

This section provides definitions of key terms to enhance understanding and clarify their usage in this study.

Back-end refers to the code that runs on the server that handles the system's core functions like the processing of attendance data, verifying facial recognition, managing events and storing of attendance data in the database.

Biometric Data refers to the facial feature data of a student used for facial registration and facial verification during event attendance confirmation.

Cancelled refers to a specific classification of event status indicating that an event has been voided or called off.

Concluded refers to a specific classification of event status indicating the phase wherein the scheduled duration of an event has ended, and the system no longer accepts new attendance entries.

Event refers to academic or non-academic event sessions organized within Rogationist College. Each event entry comprises specific event details like event name, description, time window, designated locations, and student eligibility rules.

Event Status refers to the current stage in the lifecycle of a scheduled event activity. It serves as a system trigger to determine whether certain actions such as user registration, attendance marking, or data modification are permitted or restricted at a given time.

Finalized refers to a specific classification of event status representing the terminal phase of the event lifecycle, wherein students may no longer register, and all attendance records have been audited and locked for official reporting.

Geofencing Technology refers to the technology that creates an invisible boundary around a specific event area within Rogationist College campus using GPS or the internet.

GPS (Global Positioning System) is a satellite-based navigation system that allows the student's phone to determine its precise location anywhere primarily within the Rogationist College campus.

Location refers to a manageable geofenced geographic area that is used as either a registration area where students register for an event, or an event venue where the event is conducted.

Location Environment refers to the physical setting of a location, categorized as either indoor or outdoor.

Manual Attendance Tracking is the traditional process of attendance tracking and monitoring at Rogationist College where students sign a paper to mark their attendance during school events.

Office of the Student Affairs (OSA) refers to the department that facilitates events and attendance tracking of the students.

Office of the Student Affairs (OSA) module refers to the web application used by the OSA to manage user accounts, events, locations, monitor attendance and generate attendance reports.

Ongoing is the phase indicating that an event is currently in progress and still open for late attendance registrations.

Real-time Tracking refers to the system's ability to instantly process and update attendance information as it occurs.

Registration refers to the phase during which students may officially register for an upcoming event.

Student Module refers to the mobile application used by college students to enroll in their facial biometric data, confirm their attendance at events through facial recognition and geofencing, and view event information.

Upcoming refers to the phase indicating the initial state and scheduled date and time of an event and is not yet open for any official attendance registrations.

Chapter 2

CONCEPTUAL FRAMEWORK

This chapter presents a review of related literature and studies, synthesis, and the conceptual model of the study.

Review of Related Literature and Studies

Manual attendance systems were still widely used in many schools and workplaces. Students and employees were usually required to sign logbooks or have their names checked off on printed lists. Hence, having a secure and reliable way of recording attendance was helpful to achieve the main purpose of having attendance, as well as in making it faster, so users did not have to consume time just for manually attending. According to the study of Qureshi (2025) entitled The Proposed Implementation of Radio Frequency Identification (RFID)-based Attendance System, most of the institutions, whether schools, colleges, or even companies, and workplaces around the world still use the traditional method or manual attendance systems that use printouts of attendance sheets that can be used for recording attendance within their institution. This method was associated with several problems, including being time-consuming, costly, error-prone, difficult in retrieving records, chance of loss, and being unsecured.

A manual attendance system was done either with the use of paper and pen, punch cards, or sign-in sheets to record when people arrive and leave, which caused inconvenience. Because of this, attendance monitoring over a long period of time is difficult to manage and handle. According to the study of Fernandez et

al. (2025) entitled School Event Attendance Monitoring System Using Barcode Scanner, that manually taking and keeping student attendance for a long time was hard. It wastes a lot of time and resources and could lead to mistakes. It was also not efficient, and it was difficult to manually compute the percentage of events a student has attended and record it properly.

Introducing an automated attendance system, which automated the methods of tracking and monitoring attendance with the use of geofence-based and facial recognition technology, was beneficial especially to those who will monitor it. According to Babatunde et al. (2022) with their study entitled Mobile Based Student Attendance System Using Geo-Fencing with Timing and Face Recognition, building automated attendance systems using geofence-based technology and face recognition helped to minimize manual errors and reduce incidents of attendance counterfeiting. Their system, Mobile Based Student Attendance System, used a hybrid approach, the geofencing technology to monitor the students' presence for attendance inside the geofenced perimeter and the facial recognition system to authenticate the student. The mobile applications that the students used were using GPS to detect the geo-fenced area or perimeter to monitor their presence within the needed time and location. To identify a student through facial recognition, their system uses OpenCV, which is used for real-time face detection and recognition. The system compares the student's facial data or matric number on their records stored in the database. A database was also used to store the location data of the student.

Geofencing is a location-based technology that establishes a virtual boundary around a physical geographic area. Geographic information was widely used in fields like computer software and physical security. Most existing technologies relied on GPS to accurately track a user's location, especially in outdoor settings. According to the study of Chourasia and Sargaiya (2021) entitled *Geofencing: Next Level Location Tracking Technology*, geofencing enhances tracking and monitoring systems by integrating global navigation satellite services (GNSS). These features enable organizations and businesses to obtain precise location data, thereby improving operational efficiency, security, and real-time decision-making.

Geofencing was commonly used in mobile apps to detect when a user entered or left a specific area. It relied on the device's built-in location services to detect movement, reducing the need to continuously transmit the user's exact location to a server. As noted by Shevchenko and Reips (2021) through their study entitled *Geofencing in Location-Based Behavioral Research: Methodology, Challenges, and Implementation*, that it utilized GPS, Wi-Fi, and mobile data for positioning. Android recommended a geofence radius of 100 to 150 meters, but this could be narrowed to 20 to 50 meters with Wi-Fi, and up to about five (5) meters using indoor positioning. Geofencing also operated in the background even when the app was closed, with Android supporting up to 100 geofences while iOS allowed up to 20.

Further, with the use of a geofencing automated attendance tracking system, it promotes accountability of students in attending every event in order to

avoid consequences, while at the same time being true and transparent in computing their attended events. Based on the study of Kundariya et al. (2024) entitled Geofencing and Location based Attendance System, the use of geofencing—automatic attendance marking for student's attendance promotes such transparency and accountability as this kind of attendance records in real-time, which will make students more accountable for their attendance. They stated that the use of geofencing ensured the attendance process was seamless, accurate, and efficient without the need for manual intervention. They also suggested that the use of a geofencing automated attendance tracking system had enhanced administrative efficiency by reducing the time and effort required for attendance tracking, minimizing errors, and providing educational institutions with a streamlined solution for monitoring student presence. This suggested that geofencing attendance is framed as a system that supports both administrative efficiency and student accountability.

To implement the geofencing automated attendance tracking system, web development was used as an approach for creating software applications that were accessible through web browsers on various devices. According to Parks (2022), web development is the process of creating a web application that people can access via internet browsers on different devices. Web applications are different from mobile applications that run on mobile phones and tablets. Web development is a general term that includes both front-end development and back-end development. Parks also stated that in developing web applications, web developers must have more knowledge on HyperText Markup Language (HTML),

Cascading Style Sheets (CSS), JavaScript, other programming languages, and functions of a database.

Further, this understanding is supported by Tripathi et al. (2024) on their study entitled Web Development Framework, explained explain that web development frameworks are studied using a structured approach that includes literature review and comparison of different frameworks. The study identifies front-end, back-end, and full-stack frameworks and evaluates them based on performance, usability, community support, and scalability. It also states that these frameworks are important in building efficient and scalable web applications and are tested for real-world use.

Furthermore, in line with web development, HyperText Markup Language (HTML) played a key role in building websites by organizing content using a system of tags. Mal, Shrivastava, Pandey and Mishra (2020) emphasized that HTML structures everything from headings and paragraphs to links and multimedia through a system of tags, forming the backbone of all web pages. With HTML, the language gained enhanced capabilities that improve accessibility, responsiveness, and user engagement. Thus, HTML remains essential for defining a website structure and ensuring consistent functionality across different devices and browsers.

The capabilities of HTML have further expanded with the introduction of HTML5, making it more suitable for modern web application development. Aravind et al. (2022) examined on their study entitled The Role of HTML5 and CSS3 in Creating Optimized Graphic Prototype Websites and Application Interfaces,

investigating its performance optimization techniques, cross-browser compatibility, mobile-first approaches, and accessibility considerations. Similarly, a study on HTML5 in web development noted that HTML5 supports cross-platform compatibility, designed to display web pages consistently across personal computers, tablets, smartphones, and smart televisions, while also enabling the development of complex web applications without the need for additional plug-ins. These advancements reinforce the relevance of HTML5 as a foundational technology in developing the web application for RCians Attendease, ensuring consistent performance and accessibility across different devices and browsers.

The Cascading Style Sheets 3 (CSS3) plays a vital role in shaping the visual presentation and responsiveness of modern web applications by controlling layout, styling, and user interface (UI) behavior. In their analysis of the critical role that CSS and HTML integration play in modern web development, Jain et al. (2024) highlighted that CSS3 established a modular structure in which unique modules independently specify style types, allowing for more flexible and maintainable styling across web applications. They also emphasized how CSS3 enables developers to create feature-rich interfaces that function consistently across many browsers by supporting multimedia and extensive cross-browser functionality. These developments highlight the importance of CSS3 as a fundamental styling tool in creating the RCians Attendease web application, guaranteeing responsiveness, visual coherence, and a captivating user experience across various platforms and screen sizes.

By offering strong layout and design elements appropriate for contemporary, responsive apps, CSS3's capabilities have further revolutionized web development. Parlakkiliç (2022) emphasized on their study entitled Evaluating the Effects of Responsive Design on the Usability of Academic Websites in the Pandemic, that using HTML and CSS in the main design of a website minimizes future problems, and that HTML5 and CSS3 can be used for seamless internet capability of understanding and responding to the qualities of digital devices. With CSS3, web developers gain access to advanced tools such as media queries, transitions, animations, and flexible layout systems that significantly improve how web pages are rendered across different screen sizes. Thus, CSS3 remains essential for defining a website's visual structure and ensuring a consistent, accessible experience across various devices and browsers.

While the use of TypeScript for the frontend of the web application development helped the developers as it ensured robust type safety. According to Emmanni (2021), one of the key advantages of using TypeScript is its ability to detect errors early during development. The static type system of TypeScript allows the compiler to identify type-related issues at compile time, which helps minimize errors that might otherwise occur at runtime. This feature is especially useful in large-scale applications, where tracking such issues in a dynamically typed environment can be challenging.

In support of this, the adoption of TypeScript has been supported by empirical research on software quality. Bogner and Merkel (2022) conducted a comparative analysis of code written in TypeScript and JavaScript to their study

entitled To type or not to type? A systematic comparison of the software quality of JavaScript and TypeScript applications on GitHub, with results confirming that using TypeScript enhances code readability and maintainability while reducing errors. This further validates the decision of the developers to use TypeScript in the web application, as its static typing system ensures that data type issues are caught during compilation rather than at runtime, leading to a more stable and maintainable codebase.

One of the front-end libraries used in building the web application of RCians Attendease is ReactJS. According to Bratslavsky (2025), ReactJS' widespread adoption is evident across major technology-driven industries. Industry giants like Netflix, Airbnb, and Atlassian have integrated it into their platforms to deliver fast and scalable web experiences. A key characteristic that distinguishes ReactJS from other front-end solutions is its component-based design. User interfaces are assembled from self-contained, reusable building blocks, each managing its own data and behavior. This architectural approach has contributed significantly to its popularity among developers. This is reflected in Stack Overflow's 2024 Developer Survey, which found that over two in five professional developers actively use the library. Furthermore, ReactJS follows a declarative programming model. This enables developers to focus on defining the desired state of the interface rather than manually orchestrating every change in the Document Object Model (DOM).

In relation to building a web application without the complexity of developing user interfaces and components, one front-end library that was used is ReactJS. According to the study of Aggarwal (2022) entitled Modern Web-Development

Using ReactJS, that ReactJS is an open-sourced JavaScript library that is commonly used to build interactive user-interfaces, with the purpose of creating simple, fast, and scalable web applications. Also, one of its non-complex natures is how quickly developers get comfortable with the framework. It said that the learning curve with this framework is extremely easy and gets one along without any complications. ReactJS is also known to be a highly efficient performer that makes it stand out from dozens of frameworks out there.

Further, ReactJS has demonstrated its practicality and effectiveness in the development of real-world web applications across various domains. Lazuardy and Anggraini (2022) on their study entitled Modern Front-End Web Architectures with ReactJS and NextJS, applied ReactJS in building the front-end of the State Civil Apparatus Information System of Indonesia (SIASN), a large-scale government web application used by civil servants nationwide. The study highlighted that modern web applications have become an important part of the daily ecosystem, and that choosing the right library and front-end framework plays an important role in terms of scalability, maintenance, modularity, and making web pages more interactive. In their implementation, ReactJS proved essential in optimizing the user interface, user experience, and application performance of a system deployed at a national scale. These findings affirm ReactJS' suitability not only as a development tool for simple projects but also as a dependable technology for building complex, large-scale, and user-facing web applications such as RCians Attendease.

In building the front-end of the web application, the developers used Next.js, a React-based framework designed for building powerful and scalable web applications. According to Jain and Chittezath (2023), Next.js stands out for its rendering flexibility, built-in performance optimizations, and static generation capabilities, making it a strong foundation for developing high-quality web applications. Its active developer community further contributes to its reliability and continued improvement.

A study by Susanto et al. (2023) entitled Web-Based Management Information System Design for Student Organizations in Kendal Regency Using Next.js Framework, demonstrated the practical use of Next.js in building real-world academic systems, specifically designing a web-based management information system for student organizations. Their findings highlighted how Next.js supported efficient data handling and simplified overall system design, making it a suitable framework for academic and organizational applications. This supports the adoption of Next.js in RCians Attendease as a performant and scalable front-end framework capable of handling real-time attendance data and a growing user base.

For the back end of the system, the developers used Java as the primary programming language, taking advantage of its object-oriented architecture and platform independence to build reliable and maintainable server-side applications. According to Martinez, Remegio, & Lincopinis (2023) Java is a widely used high-level, object-oriented, and class-based programming language valued for its simplicity and portability. Its simplicity, portability, and platform independence

have made it a preferred choice for developing various types of applications, including mobile, web, desktop, and games. A key advantage of Java is that it compiles into a bytecode that runs on the Java Virtual Machine (JVM), allowing applications to operate across multiple platforms without requiring system-specific modifications.

In support of this, the study of Varma (2020) entitled *The Role of Java in Modern Software Development: A Comparative Analysis with Emerging Programming Language*, Java's stability and widespread adoption in enterprise-level systems, attributing its longevity to its cross-platform compatibility through the JVM and its adherence to the write once, run anywhere principle. This supports the adoption of Java in RCians Attendease as a stable, scalable, and platform-independent language well-suited for building a reliable back-end capable of handling multiple concurrent users across different environments.

Spring Boot was chosen as the back-end development framework of RCians Attendease for its capacity to simplify Java-based application development through minimal setup and a rich collection of built-in functionalities. As discussed by Hubli and Jaiswal (2023), the Spring Boot framework has become a widely adopted solution for building reliable and production-ready backend systems at an enterprise level. The authors explain that Spring Boot is structured around the Model-View-Controller (MVC) pattern, which separates the application into distinct functional layers, each responsible for specific concerns. The Controller layer processes client requests and directs application flow, the Service layer encapsulates core business operations and rules, and the Data Access Object

(DAO) layer handles all communication with the database, covering jobs such as querying, storing, and updating data. The developers mentioned that this clear division among layers contributes to a codebase that is easier to maintain, test, and scale over time. The authors also highlight that Spring Data JPA significantly reduces the amount of repetitive code developers need to write by providing ready-made repository structures, leaving Spring to manage the underlying data access logic automatically. The framework further supports Representational State Transfer (REST) Application Programming Interface (API) development by offering dedicated annotations that define how different types of client requests are handled, with data transmitted between systems in the widely compatible JavaScript Object Notation (JSON) format. Security is also built into the framework through Spring Security, which enforces user authentication and access control to prevent unauthorized use and protects the integrity of application data. The authors get to the conclusion that Spring Boot's flexibility supports a variety of project requirements, from monolithic systems to microservice-based architectures, and that its convention-over-configuration approach lowers development complexity, allowing developers to concentrate on business logic.

Additionally, Spring Boot also allowed the system to collect data from the database and its server. According to the study of Zhang, Sun, Zheng and Dong (2021) entitled Design and Implementation of Energy Management System Based on Spring Boot Framework, that Spring Boot allows their system to include two parts on a system: the database and the server. On one hand, the database was used to store energy consumption data and user information. On the other hand,

the core component used the Spring Boot framework that was responsible for collecting, storing, and analyzing data from the data collection layer. Further, it received and processed data requests from the display interface layer as well as provided standard data interface to the external programs.

In support of this, the study of Chaturvedi (2023) entitled Modern Software Development with Java, Spring Boot, and Python: A Survey of Frameworks and Best Practices, stated that Spring Boot provides a stable and secure foundation for developing large-scale and microservice-based systems. The study highlighted that Spring Boot simplifies the configuration of Java frameworks, making it easier to use in microservice architectures and Representational State Transfer (REST) web services. It also stated that the framework supports faster development and helps in building secure and scalable enterprise applications.

In line with the design of the system, the developers used Figma as a platform to achieve a user-centered design of the system. It introduces real-time collaboration that can make the design process and prototyping of a system collaboratively effective for developers. According to an article by Ibrahim et al. (2023) they stated that Figma is a collaborative interface design tool and has garnered attention for its real time collaboration capabilities. And that Figma, a cloud-based platform, fosters teamwork among developers, allowing simultaneous editing and feedback in a shared environment.

Also, it was helpful in its features, like iterating an actual application through interactive prototypes and refining the designed interface for a better user experience. According to the study of Wijaya, Arif, Aini, and Putri (2024) entitled

UI/UX Web Based Learning Design with UCD Approach to Basic Programming using Figma, stated that Figma is a powerful platform to apply the user-centered design (UCD) to achieve a better user experience. Designers in Figma can iterate interactive prototypes and refine the interface to deliver a seamless and engaging learning experience for the target audience.

To further implement the geofencing-based automated attendance tracking system, a mobile application was developed for student use. Mobile application development is a comprehensive approach to developing software applications that are designed to operate on mobile and tablet devices. According to Hiwale, et. al. (2022), mobile application development refers to the process of creating software intended to operate on handheld devices such as smartphones and personal digital assistants, through which end users are granted access to a broad range of features and services that address their everyday needs and requirements. The authors highlighted that the use of mobile technologies has been expanding at a rapid pace, and that the presence of multiple competing platforms such as iOS and Android, each equipped with their own distinct software development kits and unique development environments, has made the overall process of building mobile software increasingly demanding and resource-intensive. The authors further outlined that successful mobile application development requires developers to have a solid command of relevant programming languages, a thorough understanding of the tools and environments used in the development process, and a working knowledge of the core components that make up a mobile application, all of which serve as essential

building blocks in producing a well-functioning and properly structured software product.

For a cross-platform mobile application without needing to develop a separate native codebase, one front-end library that was used is React Native. According to Boduch, Derks and Sakhniuk (2022), React Native is a JavaScript-based mobile app framework that allows developers to create natively rendered mobile apps for iOS and Android applications using the same code base. Therefore, the same React components that the developers build will work on any platform, even on browsers.

Unlike other mobile development frameworks, React Native stands out through its single codebase approach, allowing developers to target both iOS and Android simultaneously without maintaining separate codebases. According to the study of Balogun (2023) entitled React Native – A Native Code Integration Perspective, that React Native differs from other mobile app development frameworks in several key ways. Its single codebase nature allows developers to write code once and deploy it across both platforms, this could save time and reduce development costs and saves development time.

The practical value of React Native is further evident in its proven application across real-world, large-scale platforms used by millions of users globally. Based on the study of Prajapati and Bhattacharjee (2025) entitled React Native: A Comprehensive Analysis of Cross-Platform Mobile Application Development, conducted a comprehensive analysis of React Native through practical case studies of major technology companies. Their study found that

Facebook, the creator of React Native, reported a 30% reduction in development time by adopting the framework for features like Marketplace and Ads Manager. Similarly, Instagram integrated React Native into their existing application, enabling faster feature rollouts and consistent user experiences across platforms with minimal code duplication. Most notably, Uber Eats utilized React Native for its restaurant dashboard, which reduced development time by 40% and improved overall user satisfaction. These real-world implementations demonstrate that React Native is not merely a theoretical solution but a proven and reliable framework for building efficient, cross-platform applications, reinforcing its selection as a core technology in the development of RCians Attendease.

Expo was used as a supporting framework and toolchain to expedite the development of the React Native mobile application. According to the study of Desai (2023) entitled Comparative study of Android native vs cross-platform mobile application development, Expo's extensive toolkit, which includes pre-configured Android Studio and Xcode configurations, management of Apple and Google certificates, and push notifications, greatly streamlines working with React Native, especially for developers switching from web development. The study also noted that the Expo development team developed a tool called Snack, which lowers the barrier to entry for React Native development by enabling developers to write and test their code directly in a browser. Additionally, the developer highlighted that the Expo platform's user-friendly features and React Native's familiarity for web developers make it a desirable option for creating cross-platform mobile applications. These features, which enable effective and accessible mobile

application development across both iOS and Android platforms, support the choice of Expo as the development framework for RCians Attendease.

Facial recognition was one of the features of the mobile application that enhanced user convenience and security by verifying a user's identity during login and protecting user data from unauthorized access. Andrejevic and Selwyn (2020) stated that facial recognition promises conveniences to users for faster transactions with enhanced security. They discussed how promising it is to put an end to the inevitable gaps that arise when teachers are tasked with repeatedly conducting roll-calls of large student groups in monitoring the students' attendance. They also stated that facial recognition applications promise multiple benefits and conveniences including speedy and secure transactions, customized services, and enhanced public safety and security. This suggested that facial recognition is framed as a multifunctional technology that supports both operational efficiency and enhanced security.

In support of this, a study of Al-Muhaidhri and Hussain (2019) entitled Smart Attendance System Using Face Recognition, wherein they developed a facial recognition-based attendance system that automatically identified students and recorded their attendance without manual intervention. The study found that the system improved efficiency by reducing human error and eliminating the need for traditional roll-call methods. It also demonstrated that facial recognition can provide a faster and more reliable way of monitoring attendance in educational environments.

Facial recognition within RCians Attendease was implemented using OpenCV. OpenCV is a widely used image processing library, particularly for facial recognition applications. According to Gaikwad (2023), OpenCV is an open-source computer vision and machine learning software library that has achieved broad adoption among developers and researchers worldwide. The author further noted that OpenCV provides extensive functionalities essential for vision-based systems, such as image and video processing, object detection and tracking, facial recognition, and machine learning integration. Additionally, OpenCV is designed to be cross-platform, operating on major operating systems including Windows, Linux, macOS, Android, and iOS. It supports multiple programming languages, such as C++, Python, Java, and MATLAB, which enhances its accessibility and adaptability across diverse development environments.

One of the key factors that led the developers of the RCians Attendease to implement the OpenCV library to the system is its speed, portability, and cost-effectiveness. According to the study of Anand et al. (2021) entitled Facial Recognition System Using OpenCV, that OpenCV provides advanced proficiencies like face detection, face tracking, facial recognition, and many more methods for artificial intelligence (AI) which the users can assure that the image processing capabilities of this would highly be trusted and helpful for systems that is a facial recognition-based. OpenCV also offers free and open-source nature which allows developers to integrate facial recognition solutions without any cost because of its minimal license. It also uses C/C++ library functions which would

help in faster execution in image processing and implements on C, therefore any devices which run on C can run OpenCV.

The effectiveness of OpenCV in automating attendance through facial recognition has been demonstrated in related studies. Ali et al. (2021) through their study entitled Automatic Face Recognition Attendance System Using Python and OpenCV, developed an automatic face recognition attendance system using Python and OpenCV, employing a Histogram of Oriented Gradients (HOG) for face detection and a Support Vector Machine (SVM) classifier for recognition. The system achieved a 99.38% accuracy rate on the Labelled Faces in the Wild benchmark, confirming the reliability of OpenCV-based facial recognition in academic settings. This supports the adoption of OpenCV in RCians Attendease as a proven and accurate approach to automating student attendance verifications.

One of the programming languages used in developing the facial recognition service of RCians Attendease is Python. According to the article of Dhruv, Patel, and Doshi (2021), Python is a programming language that follows an object-oriented, interpreted, and interactive design, offering developers access to advanced data structures, dynamic typing and binding, modular programming support, and built-in mechanisms for automatic memory management. The article further noted that Python can support parallel computing environments and is characterized by a straightforward and readable coding syntax, which together make it both a powerful tool and an approachable option for developers across different skill levels. The authors also pointed out that Python's rich collection of pre-built libraries significantly cuts down the volume of code that developers are

required to produce, a quality that has driven its increasing relevance across domains including machine learning, data science, image processing, and deep learning. One of the image processing libraries covered in the article is Pillow, identified as a Python Imaging Library that broadens the language's built-in capabilities by enabling applications to open, process, and store images in numerous file formats, making it a valuable tool for systems that rely on handling visual data such as facial images.

In support of this, the study of Sain, Dutta, Saha and Mandal (2023) entitled Automated Facial Recognition based Attendance System using OpenCV in Python, validated the effectiveness of Python as a programming language for real-time facial recognition by developing and deploying an attendance system that combined Python with OpenCV across standalone devices. The system underwent testing with participants from various age groups and genders, consistently producing accurate identification results while automatically logging each individual's attendance together with the corresponding time of detection. The developers concluded that transitioning from conventional manual attendance methods to a Python-driven facial recognition approach significantly reduces the occurrence of recording inaccuracies, overlooked entries, and repeated submissions, thereby establishing Python as a dependable and practical foundation for building image-based identification systems like RCians Attendease.

The developers used IntelliJ IDEA as the integrated development environment (IDE) for constructing the back-end component of RCians

Attendease. According to the article of Magar, et al. (2023), IntelliJ IDEA stands as one of the most extensively adopted development environments available, recognized for its capacity to meaningfully boost the efficiency and output of software developers. The article characterized IntelliJ IDEA as a platform equipped with smart coding support capabilities, offering built-in functionalities such as context-sensitive code recommendations, automatic code completion, and on-the-fly error identification, all of which contribute to a faster and more precise coding process. The authors further highlighted that the environment is built to readily support the addition of external tools and extensions, giving it the flexibility to function effectively across a wide range of programming languages and project configurations. The article also reported that IntelliJ IDEA's language compatibility was put to the test across several major programming languages including Java, Python, and JavaScript, with the environment achieving an 85% success rate in correctly interpreting and producing code in these languages, underscoring its broad and dependable language coverage. Given this versatility, and most notably its strong compatibility with Java, IntelliJ IDEA proved to be a well-suited development workspace for RCians Attendease, equipping the development team with the necessary tools to efficiently construct, evaluate, and enhance the Java-driven back-end operations of the system.

In support of this, a study of Kurbatova, Golubev, Kovalenko and Bryksin (2021) entitled *The IntelliJ Platform: A Framework for Building Plugins and Mining Software Data*, stated that IntelliJ offers a wide range of powerful features that can greatly benefit the developers. These features support the development of various

tools and the implementation of diverse methodologies, making it an ideal foundation for creating advanced software engineering solutions and conducting experimental research in the field.

Another IDE that the developers used was Visual Studio Code. According to an article by Industry Wired (2024), Visual Studio Code (VSCode), continues to be the most popular code editor for developers in 2024. Its lightweight build, quick responsiveness, and extensive customization make it stand out. Visual Studio Code accommodates various programming languages like Python, JavaScript, and TypeScript through a rich extensions marketplace, and Developers can also tailor the editor to their workflow with customizable themes, plugins, and settings.

In support of this, the study of Tan, Chen, and Jiao (2023) entitled Visual Studio Code in Introductory Computer Science Course: An Experience Report validated Visual Studio Code as a development environment that operates across multiple platforms, offers a gentle learning curve, and comes equipped with a broad range of built-in capabilities that have made it a preferred tool among professional developers. The study emphasized that development environments like Visual Studio Code contribute meaningfully to the programming workflow by recognizing language syntax and offering integrated tools for tasks such as automating builds, checking code quality, running tests, and identifying errors, all of which work together to make the development process faster and less burdensome for programmers. The developers also observed that developers who had prior exposure to industry settings were particularly vocal about preferring

Visual Studio Code over alternative development platforms, citing its widespread use in real-world professional environments as a key reason for their preference, which further establishes it as a fitting and practical choice for developing the front-end components of RCians Attendease.

The developers selected MongoDB as the database management system responsible for storing both user information and geospatial data in RCians Attendease. According to the article of Rathore and Bagui (2024) explained that MongoDB has undergone continuous refinement ever since it was first introduced in 2009, with each successive version bringing enhancements in areas such as processing speed, system scalability, and data protection. The authors characterize MongoDB as a document-oriented database that organizes information into adaptable, JSON-like documents grouped within collections, a design that moves away from the fixed, table-based structure associated with conventional relational databases and makes it significantly easier to accommodate varied and frequently changing data formats. One of the most relevant aspects of MongoDB for this system is its built-in capacity to handle geospatial information. The authors pointed out that MongoDB's architecture natively incorporates geospatial functionality, enabling applications to store, retrieve, and perform spatial analysis on location-based data without relying on external tools or extensions. The database accommodates a variety of geographic representations, such as points, lines, polygonal boundaries, and multi-dimensional shapes, and pairs these with dedicated spatial indexing that significantly speeds up the execution of location-based queries. This makes

MongoDB particularly well-suited for applications centered around geographic awareness, including systems that perform map-based operations. The developers further emphasized that MongoDB's combined support for geospatial queries and complex data structures makes it a favored solution in fields that process large volumes of location-sensitive information such as location-based service platforms.

Supporting this, according to the study of Chauhan (2019) entitled *A Review on Various Aspects of MongoDB Databases*, stated that MongoDB is a leading open-source document-oriented database that provides high-performance data persistence and a rich query language supporting create, read, update, and delete operations. The study also demonstrated that MongoDB delivers lower execution times compared to other databases across all four (4) basic operations, making it well-suited for applications that need to support thousands of simultaneous users. Additionally, MongoDB supports horizontal scalability through sharding, which distributes data across multiple machines to improve both performance and availability.

Managing the integrity and history of the codebase was essential to the development process. For this purpose, the developers used Git as the version control system to collaborate effectively throughout the software development lifecycle. Based on the article of Worsley (2023), Git allows users to work individually on projects locally, save any changes that work, then synchronize those changes to a Git repository so others can see newer version and allows numerous team members to work collaboratively on a project. It also tracks and

records changes to any file, which makes the developers recall specific iterations later or as needed.

In support of this, the study of Ghodke and Chavan (2024) entitled An Overview of Git, stated that Git is widely adopted in both academic and industrial software projects, enhancing collaboration, improving project management, and supporting accountability among developers. Their study further emphasized Git's educational value, noting that it prepares students for real-world development practices. This supports the adoption of Git in RCians Attendease as a foundational tool for maintaining code integrity and coordinating development across the team.

The developers adopted GitHub as the primary platform for version control and collaboration, given its support for open-source repositories that make code accessible to contributors and team members alike. According to Rubei, et al. (2020), GitHub is one of the most used development services that includes version control systems and made it a popular platform that aggregate the open-source repositories feature that supports collaborative development activity such as bug tracking, contribution requests, task management, and wikis.

Furthermore, a study of Tiwari, Rathore, Farooq, Patani (2022) entitled How Students Are Using Github? A Survey emphasized that GitHub promotes developer collaboration and provides visibility into team activity on shared projects. This supports the adoption of GitHub in RCians Attendease as a reliable platform for managing source code, tracking development progress, and ensuring transparency across the development team.

The backend services of RCians Attendease using Amazon Web Services (AWS) Elastic Compute Cloud (EC2) to host the application in a scalable cloud environment. Based on the article of Reddy (2024), a Spring Boot application can be deployed to an EC2 instance through the Command Line Interface (CLI) by packaging the application into a Java Archive (JAR) file, provisioning a cloud instance, and running it on the remote server, making cloud deployment accessible even for small development teams.

In support of this, Kewate, et al. (2022) described AWS EC2 as a service that functions as a virtually unlimited set of virtual machines, offering various configurations of CPU, memory, storage, and networking resources to meet different workload requirements. This confirms the suitability of EC2 as a deployment platform for the RCians Attendease backend services.

Synthesis

Through the reviewed related studies and literature, the researchers conceptualized the study and considered those ideas to develop an attendance monitoring system using geofencing and facial recognition technologies for the College Department of Rogationist College.

The idea of automating the manual monitoring of attendance is similar to the study of Babatunde et al. (2022) wherein they use Geofencing to record the student's attendance only if the student is within a specific geographical area, and Face Recognition for verification of users' identity which aims to reduce issues of manual recording of attendance, improve accuracy and reduce fraud. Additionally,

their system also uses a mobile-based implementation approach for easier monitoring of the attendance. While for the development of the overall system, the developers had gathered more studies to determine which technology stack to use. For the backend, the developers used Java, a high-level, object-oriented, and class-based programming language, as noted by Martinez, Remegio, & Lincopinis (2023). To complement Java, the developers adopted Spring Boot as the back-end development framework for its capacity to simplify Java-based application development through a stable and secure foundation for developing large-scale and microservice-based systems as supported by the study of Chaturvedi (2023). For building the frontend of the web application, ReactJS library was used, the developers considered the study of Lazuardy and Anggraini, (2021) noting that using ReactJS library components make it easy for developers to build UI components by separating it into several parts, the components, so that the design and process of building the UI will be effective and well structured. To complement ReactJS, the developers adopted Next.js as its supporting framework. The decision of using Next.js was supported by Susanto et al. (2023), demonstrated the practical use of Next.js in building real-world academic systems, highlighting how it supported efficient data handling and simplified overall system design. For the mobile application, the developers used React Native, as supported by Boduch, Derks, and Sakhniuk (2022), using React Native enables developers to create natively rendered mobile applications from a single codebase allowing RCians Attendease to reach students across both iOS and Android platforms without doubling development effort. For the facial recognition service specifically,

the developers utilized OpenCV, an open-source computer vision library. Using OpenCV library is supported by the findings of Ali et al. (2021), whose OpenCV-based facial recognition system achieved a 99.38% accuracy rate in academic settings, confirming its reliability for student attendance verification in RCians Attendease. The developers used Python to implement OpenCV facial recognition service, this decision is supported by the study of Sain, Dutta, Saha and Mandal (2023) noting that Python-driven facial recognition approach significantly reduces the occurrence of recording inaccuracies, overlooked entries, and repeated submissions, thereby establishing Python as a dependable and practical foundation for building image-based identification systems. On one hand, for the IDE's, the developers used Visual Studio Code for the development of front-end components, according to the study of Tan, Chen, and Jiao (2023) it offers a gentle learning curve, and comes equipped with a broad range of built-in capabilities which work together to make the development process faster and less burdensome for programmers. On the other hand, for the development of the back-end component, the developers used IntelliJ IDEA, as supported by the study of Kurbatova, Golubev, Kovalenko and Bryksin (2021) it supports the development of various tools and the implementation of diverse methodologies, making it an ideal foundation for creating advanced software engineering solutions and conducting experimental research in the field. While for the prototyping of user interface and user experience (UI/UX) of the web and mobile application, the developers used Figma because according to the study of Wijaya, Arif, Aini, and Putri (2024) it is a powerful platform to apply the user-centered design (UCD) to

achieve a better user experience that designers can iterate interactive prototypes and refine the interface to deliver a seamless and engaging learning experience for the target audience. For the database management system of RCians Attendease, the developers used MongoDB to handle the storage of both user information and geospatial data. Rathore and Bagui (2024) described MongoDB as a document-oriented database that organizes information into flexible, JSON-like documents grouped within collections. Particularly relevant to RCians Attendease is MongoDB's built-in geospatial functionality, which enables applications to store, retrieve, and perform spatial analysis on location-based data without relying on external tools, supporting geographic representations such as points, lines, and polygonal boundaries paired with dedicated spatial indexing that speeds up location-based queries. While for version control, the developers used Git to collaborate effectively throughout the software development lifecycle, supported with the study of Ghodke and Chavan (2024) that it enhances collaboration, improving project management, and supporting accountability among developers, which makes the developers considered it as the foundational tool for maintaining code integrity and coordinating development across the team. The developers also adopted GitHub as the primary platform for version control, given its support for open-source repositories that make code accessible to contributors and team members alike which was supported by the study of Tiwari, Rathore, Farooq, Patani (2022). And lastly, for the AWS EC2 the developers had considered it as it was described as a service that functions as a virtually unlimited set of virtual machines, offering various configurations of CPU,

memory, storage, and networking resources to meet different workload requirements as supported through the study of Kewate, et al. (2022).

While both this study and the work of Babatunde et al. (2022) share the common goal of improving attendance monitoring through automation, they differ in scope, purpose, and technical implementation. Babatunde et al. (2022) developed their system specifically for classroom attendance at Kwara State University, utilizing Firebase as their document-oriented real-time database platform and GeoFire as a complementary library for real-time geolocation querying within Firebase. In contrast, RCians Attendease is designed for school event attendance at Rogationist College, which carries a different set of requirements compared to routine classroom monitoring. Additionally, the developers opted for MongoDB over Firebase after encountering compatibility issues when integrating Firebase with Spring Boot, making MongoDB the more practical and seamless choice for the system's back-end architecture. Also, the developers did not use Android Studio as the IDE for the development of the mobile application, instead the developers used Visual Studio Code. While other studies like Qureshi (2025), which utilized a RFID-based system for student tracking, and Fernandez (2025), which employed barcode scanners for attendance monitoring, RCians Attendease uses a geofencing-based system to automate student attendance tracking. Despite these differences in context and technology, both systems converge on the same fundamental objective which is eliminating the inefficiencies of manual attendance tracking. By combining geofencing and facial recognition, RCians Attendease ensures that students are

not only physically present within the designated event area but also that their identities are accurately verified, resulting in a secure, automated, and reliable attendance tracking solution for college students.

Conceptual Model of the Study

The conceptual model of the study in Figure 1 presents the system requirements, processes involved in constructing the system, and output of the study. It is composed of four (4) blocks: input, process, output, and evaluation blocks.

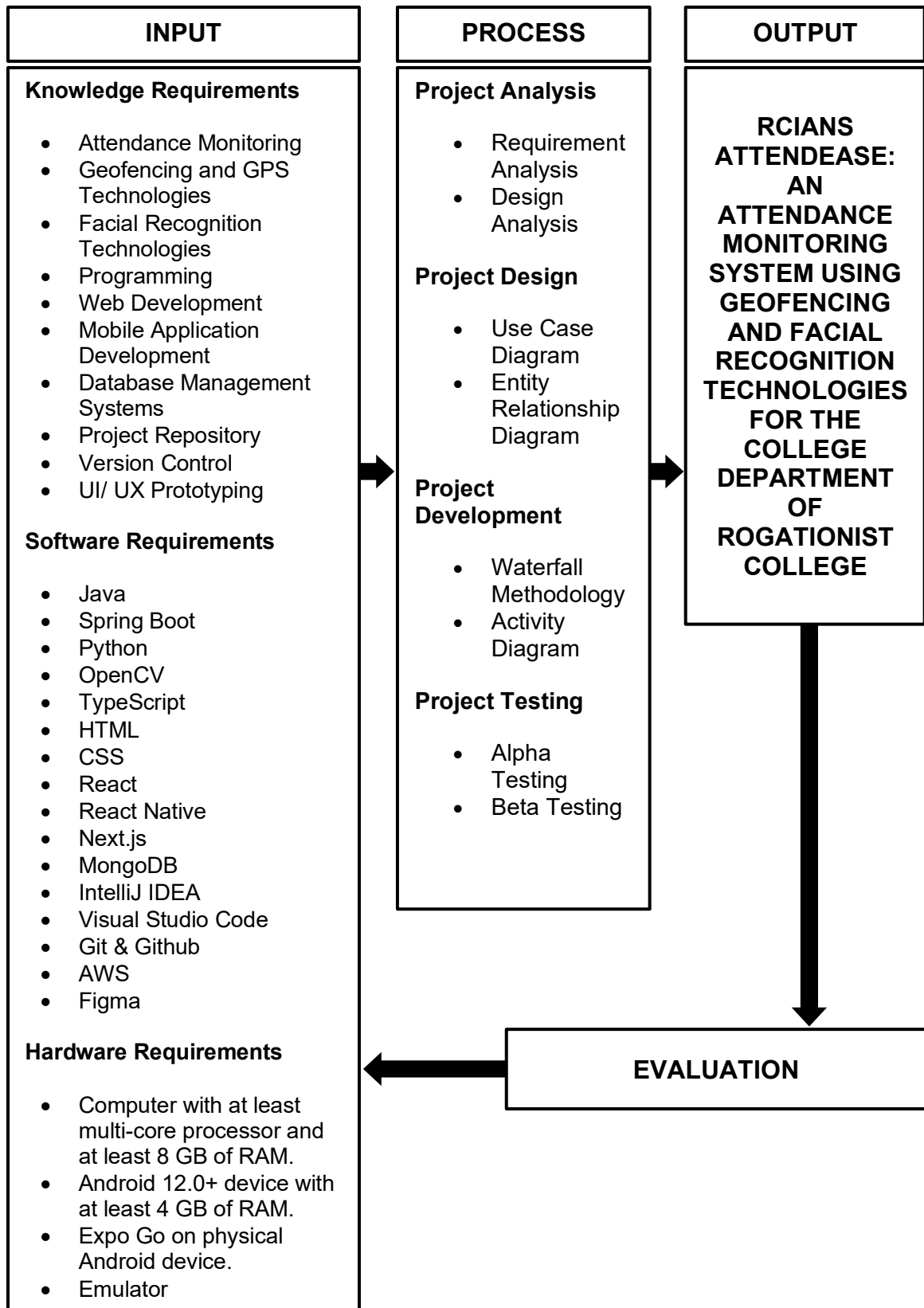


Figure 1. Conceptual Model of the Study

The input block includes knowledge requirements, software requirements and hardware requirements. In knowledge requirements, knowledge about attendance monitoring, geofencing technologies, facial recognition technologies, programming, web development, mobile application development, database management systems, project repository, version control, and user interface/ user experience (UX) prototyping are included. For the development of the system, software applications are needed. These include programs like Java, Spring Boot, Python, OpenCV, TypeScript, HTML, CSS, React JS, React Native, Next.js, MongoDB, IntelliJ IDEA, Visual Studio Code, Git and Github, and Figma. Physical materials are also required in system development, these are the hardware requirements which include a computer with at least multi-core processor with 8GB of RAM, Android 12.0+ with at least 4GB of RAM, Expo Go on physical Android device and an Emulator.

The process block refers to the process involved during the development of the system. It includes project analysis, project design, project development and testing. Analyzing the design and functional requirements of the system is important to know what the system is all about. Project design includes the construction of Use Case Diagram and Entity Relationship Diagram. In the project development, the Software Development Life Cycle with Waterfall Methodology and Activity Diagram are used in determining the processes in the system. After development, the system underwent alpha testing conducted by the developers and advisers, followed by beta testing with students and the OSA coordinator.

Finishing the said processes resulted in the completion of the modules and of the entire system.

After the output block, evaluation follows which is used in determining if the system meets the objectives set into it. Analysis of the result of the evaluation was used for improvements in the input phase again.

Chapter 3

RESEARCH METHODOLOGY

This chapter presents project design, project development, operation and testing procedure, and evaluation procedure.

Project Design

This study aimed to develop an automated attendance tracking and monitoring system for the College Department of Rogationist College. The system consists of two (2) interconnected modules: the OSA Module and the Student Module. The OSA Module is accessible via a web application, while the Student Module is accessible via a mobile application.

To access the web application, the OSA coordinator logs into the system to manage student accounts, events, and venues, as well as monitor ongoing events and view event histories and attendee records. Meanwhile, students access the mobile application by logging in with their school account, which allows them to register for events and track their attendance.

Figure 2 presents the Use Case Diagram of the RCians Attendease

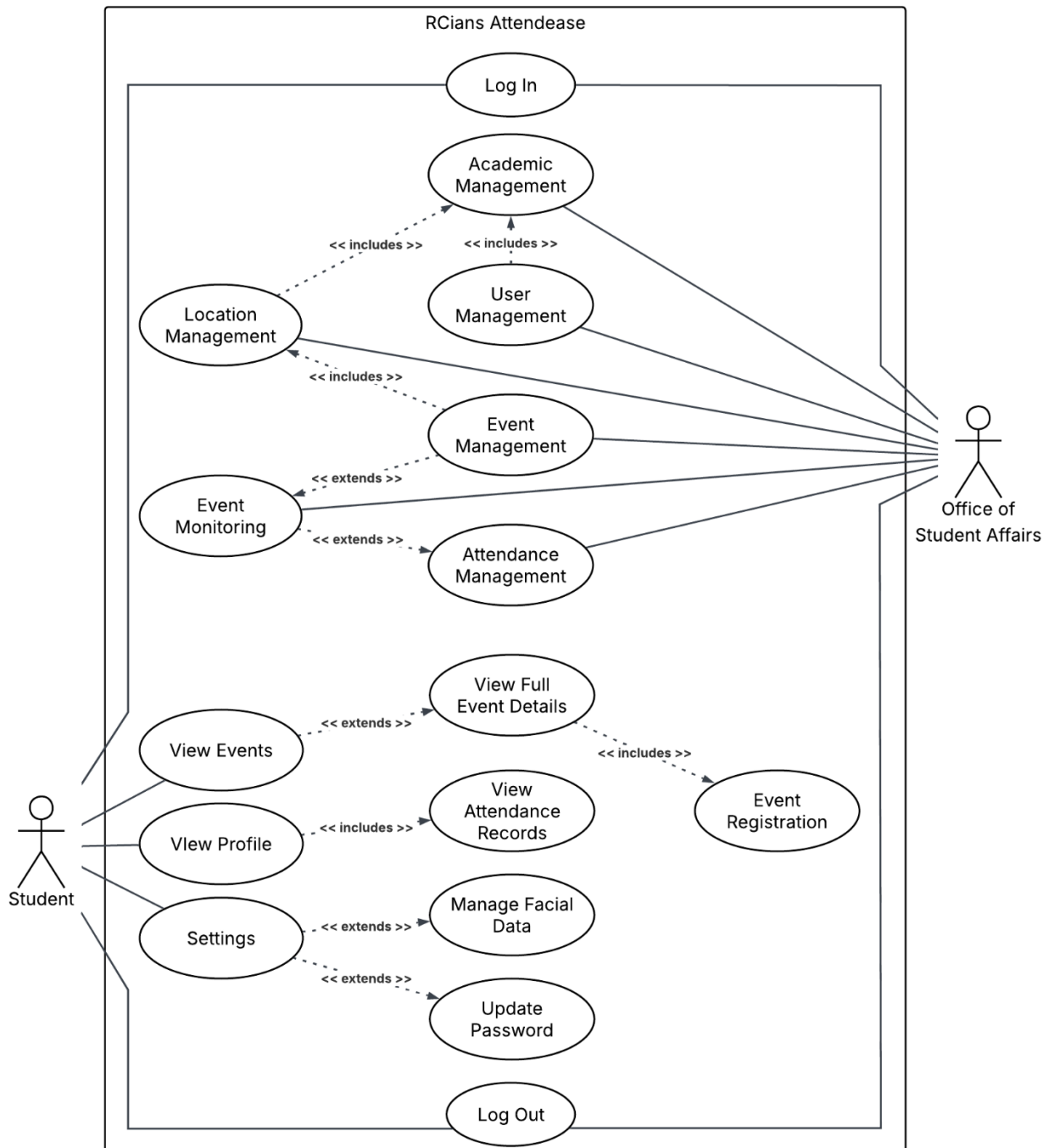


Figure 2. Use Case Diagram of the RCians Attendease

In the figure, the OSA coordinator is responsible for managing academic details including academic years, semesters, clusters, courses, and sections, as well as creating and maintaining student accounts with the required basic information. The OSA coordinator also handles event management, which covers

creating and updating events and their details, monitoring live attendance by tracking and filtering the attendee list, and reviewing generated records and analytics from concluded events for counter-checking purposes.

Students, on the other hand, can log in and out of the system, complete a one-time facial registration during initial setup, view events currently listed under upcoming, registration, and ongoing statuses, and view their account information and attendance records.

Project Development

The developers adopted the Waterfall Model of the Software Development Life Cycle (SDLC), which is a sequential approach wherein each phase must be fully completed before the next one begins. As Arya (2024) noted, it can be highly effective for projects with clear, stable requirements and where a linear approach is advantageous. She also identified the model's core phases as requirements, system design, implementation, testing, deployment, and maintenance, as illustrated in Figure 3.

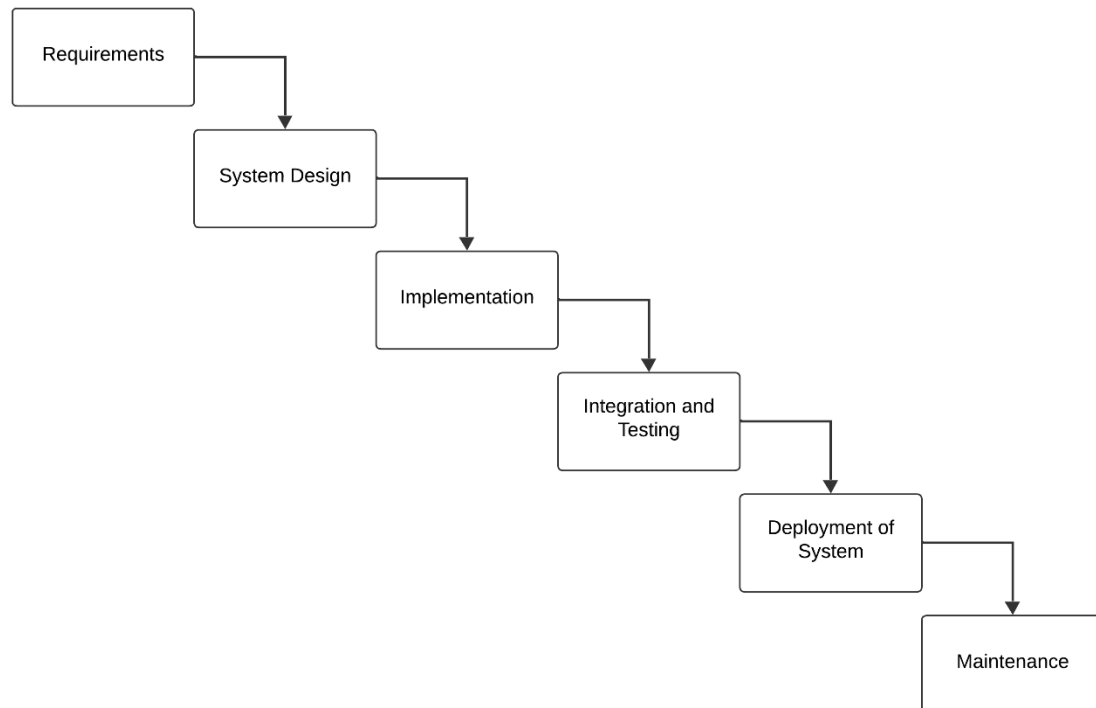


Figure 3. Waterfall Methodology

The Waterfall Model provides a structured framework for guiding software development by organizing the process into distinct phases: planning, design, implementation, integration and testing, system deployment, and maintenance. Each phase is carried out sequentially, meaning a new phase only begins once the preceding one has been completed.

Requirement Phase

To establish the project's goals, the developers gathered system requirements through research, focusing on the needs of the Office of Student Affairs and the challenges associated with tracking student attendance during events, based on a personal communication conducted with Sempio (2025). These requirements guided the planning of the system and ensured that key features

such as real-time attendance tracking, facial recognition, geofencing, and event reporting were incorporated based on both the identified needs and the ideas contributed by the researchers.

The concept of developing an automated attendance system incorporating geofencing and facial recognition emerged from the researchers' review of existing studies, which highlighted the limitations of manual attendance monitoring. The system aimed to address these shortcomings by making the attendance monitoring process digitalized, faster, more efficient, and more accurate.

System Design Phase

The overall software architecture of the RCians Attendease was determined by drawing from the related studies of similar systems reviewed in prior discussions. From these references, the developers were able to define and translate design concepts into a comprehensive blueprint for the system, covering its software architecture, components, interfaces, and data flow.

Figure 4 presents the system architecture of RCians Attendease.

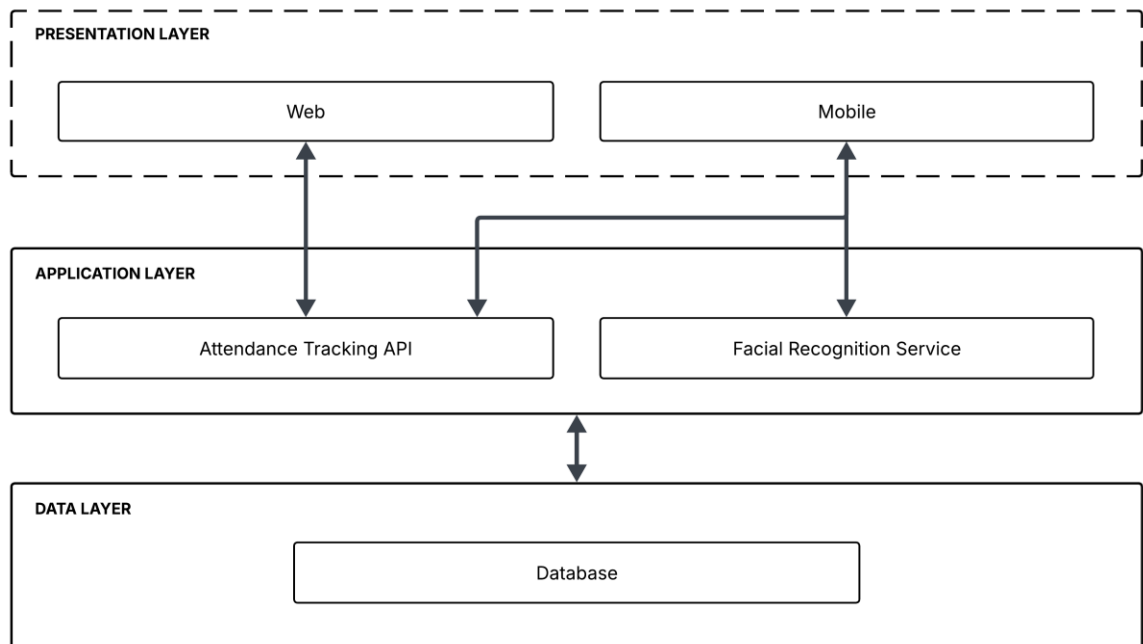


Figure 4. Software Architecture of RCians Attendease.

The figure illustrates the system architecture of RCians Attendease built upon the Three-Tier Architecture; organizing layers into three logical tiers: the Presentation Layer, Application Layer, and Data Layer.

The presentation layer serves as the system's frontend and is responsible for handling user interactions as well as displaying and collecting information. The presentation layer of RCians Attendease consists of two (2) interfaces: the OSA accesses the system via a web application, while students interact with it through a mobile application.

The application layer functions as the system's backend, where all business logic is executed. Requests received from the presentation layer are processed according to predefined business rules. The layer is composed of two services: the Attendance Tracking API, built with Spring Boot, which serves as the core backend of the system responsible for authentication, attendance tracking, geofencing,

event monitoring, event management, user management, location management, and academic data management; and the Facial Recognition Service, built with Python, which manages the facial recognition pipeline as an independent service. The Attendance Tracking API delegates facial recognition tasks to the Facial Recognition Service via REST, and interfaces with the Data Layer to perform database operations.

The Data Layer serves as the system's database layer, where all information processed by the application layer is stored and managed. The Attendance data layer is handled using MongoDB as its database management system.

Figure 5 presents the activity diagram of the OSA module in authentication and navigation flow.

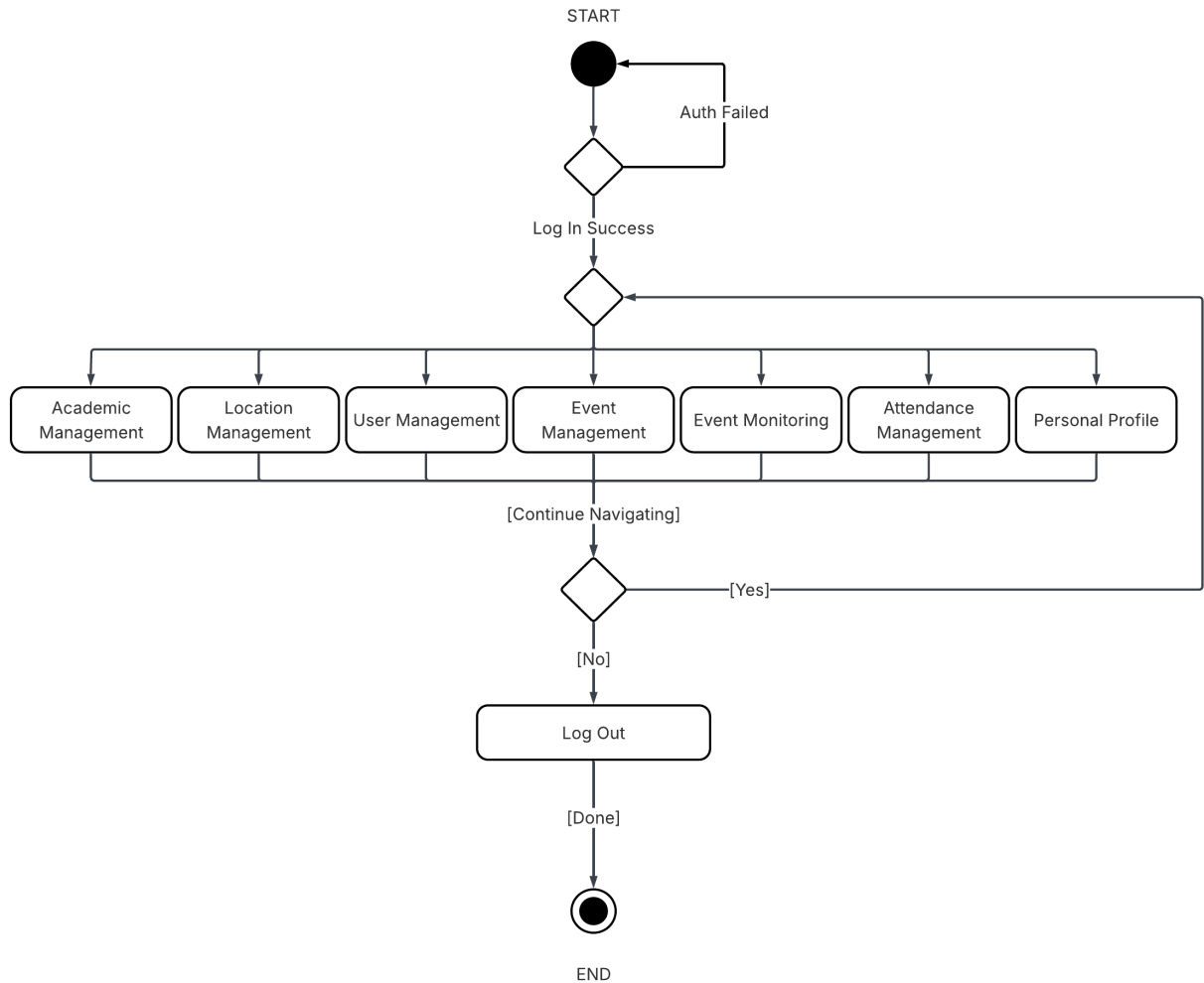


Figure 5. Activity Diagram of the OSA Module Authentication and Navigation Flow

The process begins when the user attempts to log in; if authentication fails, the system redirects the user back to the login to retry. Upon successful authentication, the user is granted access to seven (7) modules: Academic Management, Location Management, User Management, Event Management, Event Monitoring, Attendance Management, and Personal Profile. The user may continue navigating between modules as needed. Once the user chooses to end the session, the system proceeds to log out, terminating the process.

Figure 6 presents the activity diagram of the OSA module in academic management.

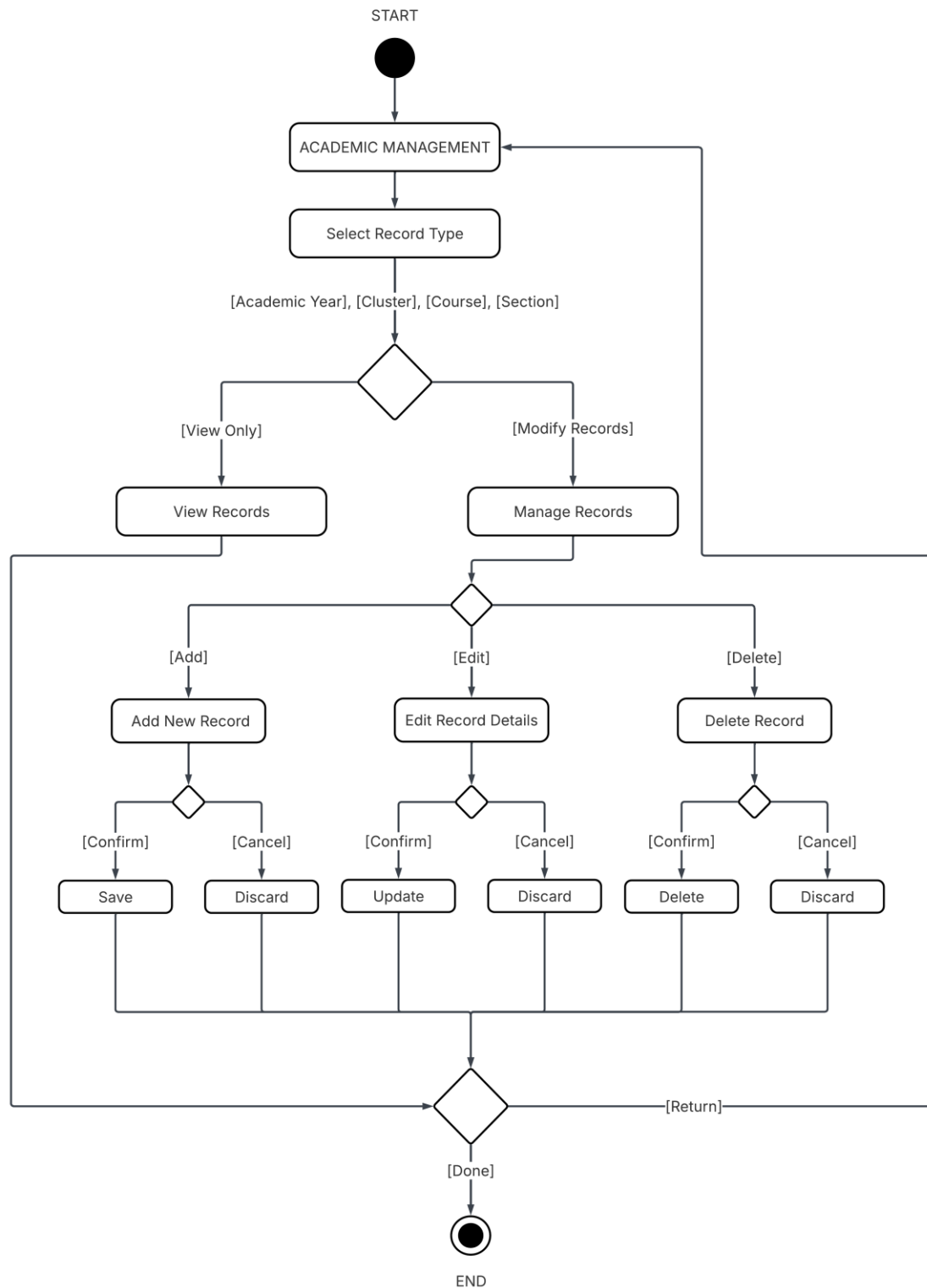


Figure 6. Activity Diagram of the OSA Module in Academic Management

The figure illustrates the academic data management process, which covers four (4) record types: academic year, cluster, course, and section. The OSA coordinator first selects a record type, then chooses to either view or modify records. When modifying, the coordinator may add, edit, or delete a record, each requiring confirmation before the change is saved or discarded. After completing an operation, the coordinator may return to manage another record or exit the process.

Figure 7 presents the activity diagram of the OSA module in Location Management.

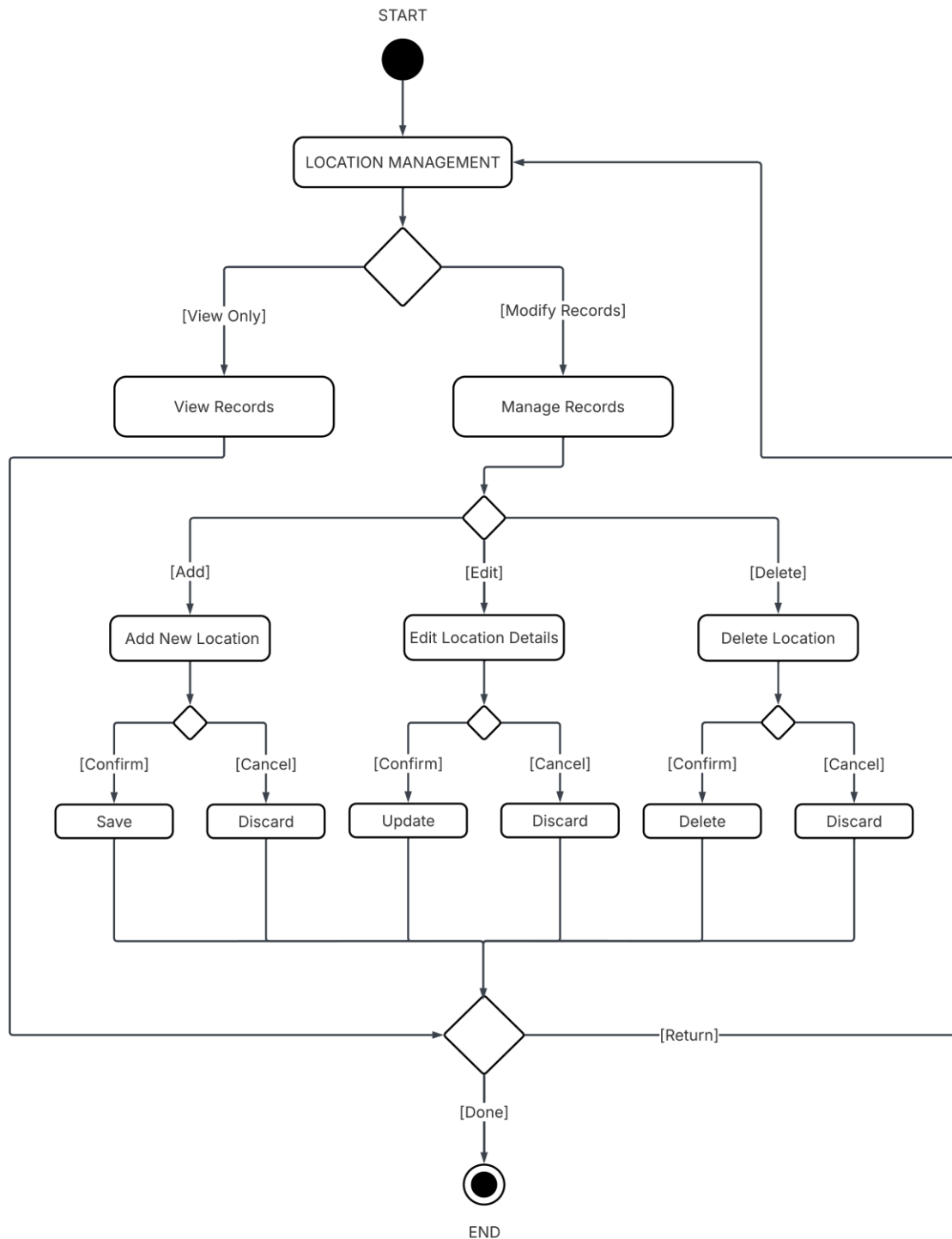


Figure 7. Activity Diagram of the OSA Module in Location Management

The figure illustrates the location management process, where the OSA coordinator may either view or modify location records. When modifying, the coordinator may add a new location, edit an existing location's details, or delete a selected location, each requiring confirmation before the change is saved or discarded. After completing an operation, the coordinator may return to managing another location or exit the process.

Figure 8 presents the activity diagram of the OSA module in user management.

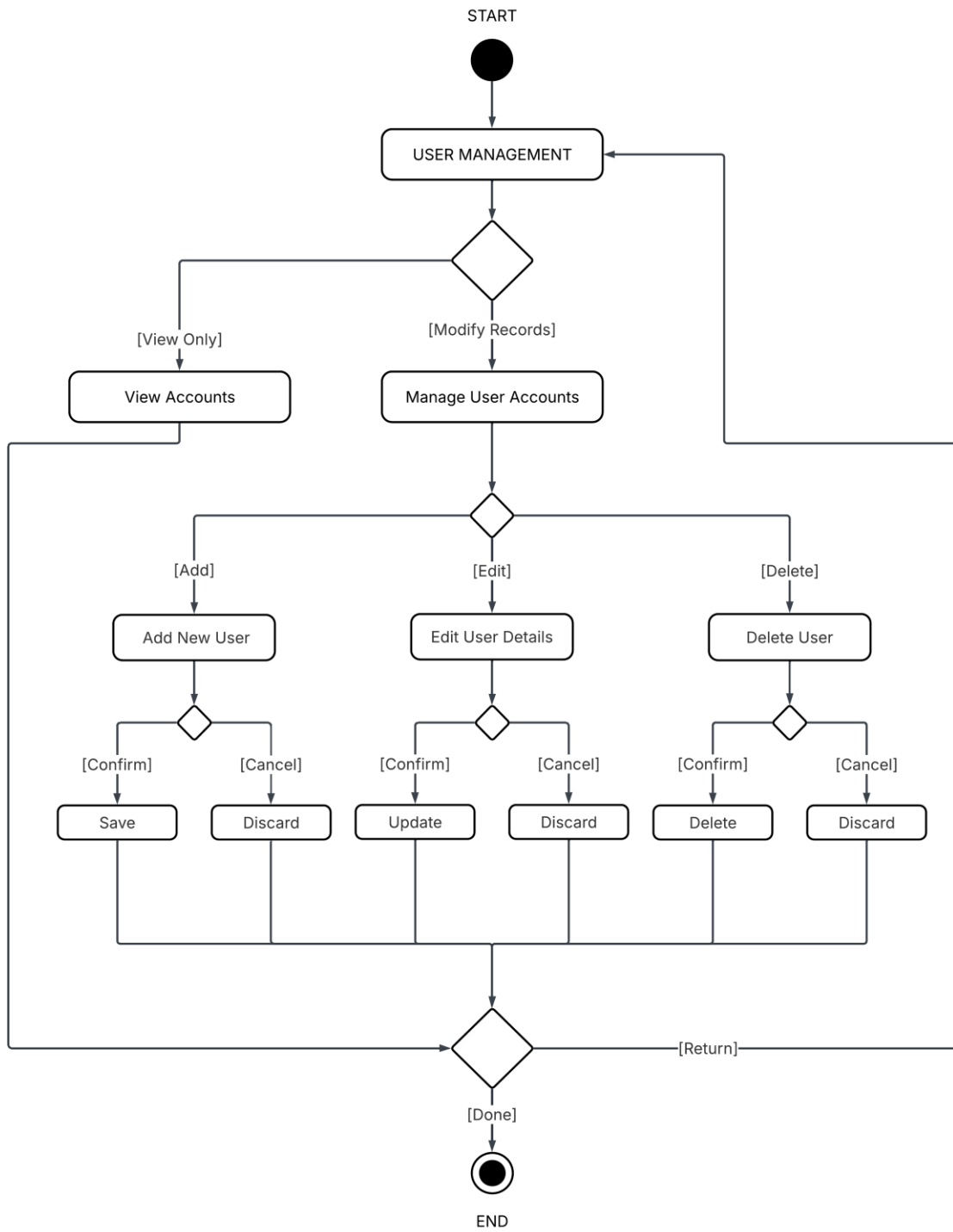


Figure 8. Activity Diagram of the OSA Module in User Management

The figure illustrates the user management process, where the OSA coordinator may either view or modify user accounts. When modifying, the coordinator may add a new user, edit an existing user's details, or delete a selected user, each requiring confirmation before the change is saved or discarded. After completing an operation, the coordinator may return to managing another account or exit the process.

Figure 9 presents the activity diagram of the OSA module in event management.

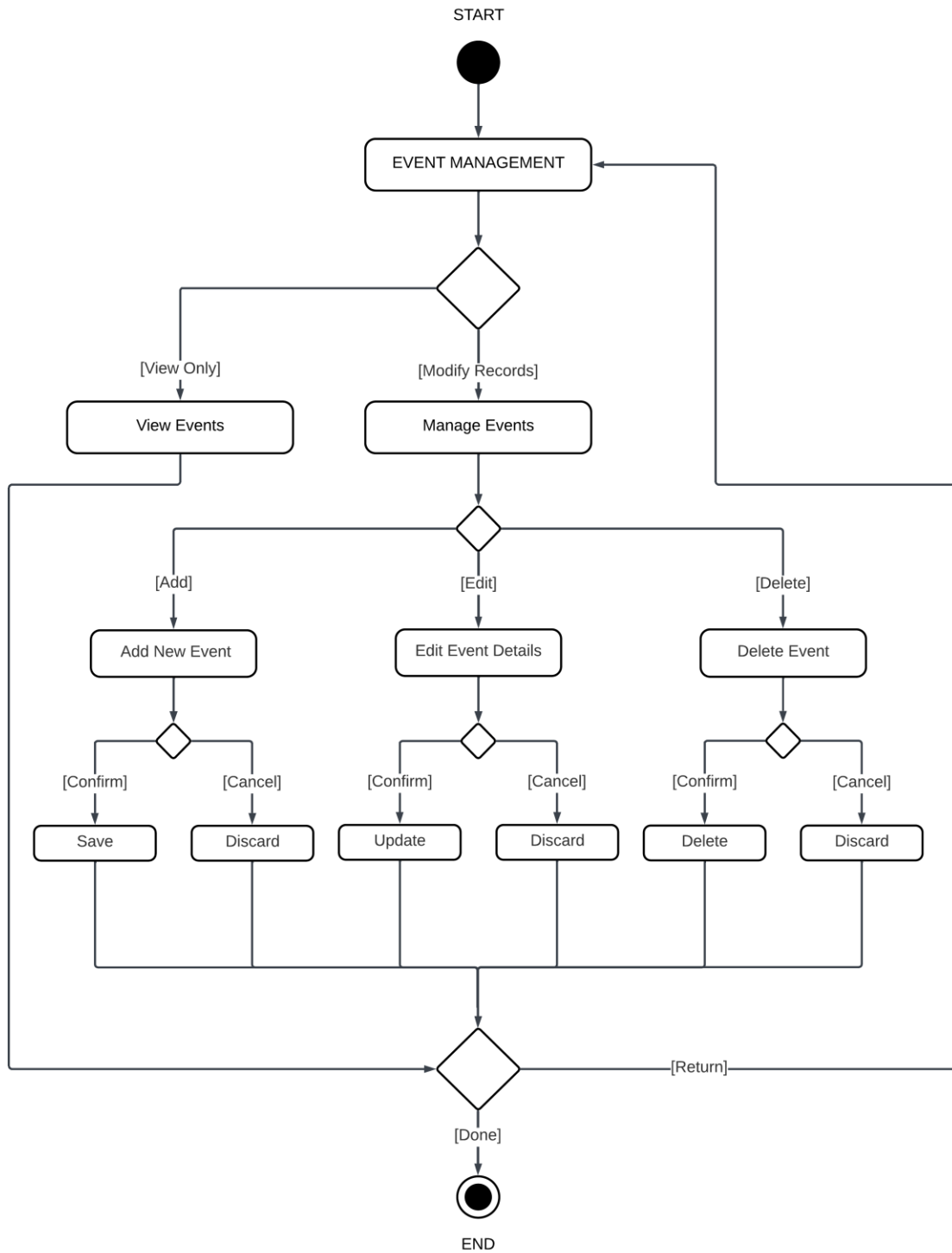


Figure 9. Activity Diagram of the OSA Module in Event Management

The figure illustrates the event management process, where the OSA coordinator may either view or modify events. When modifying, the coordinator may add a new event, edit an existing event's details, or delete a selected event, each requiring confirmation before the change is saved or discarded. After completing an operation, the coordinator may return to managing another event or exit the process.

Figure 10 presents the activity diagram of the OSA module in event monitoring.

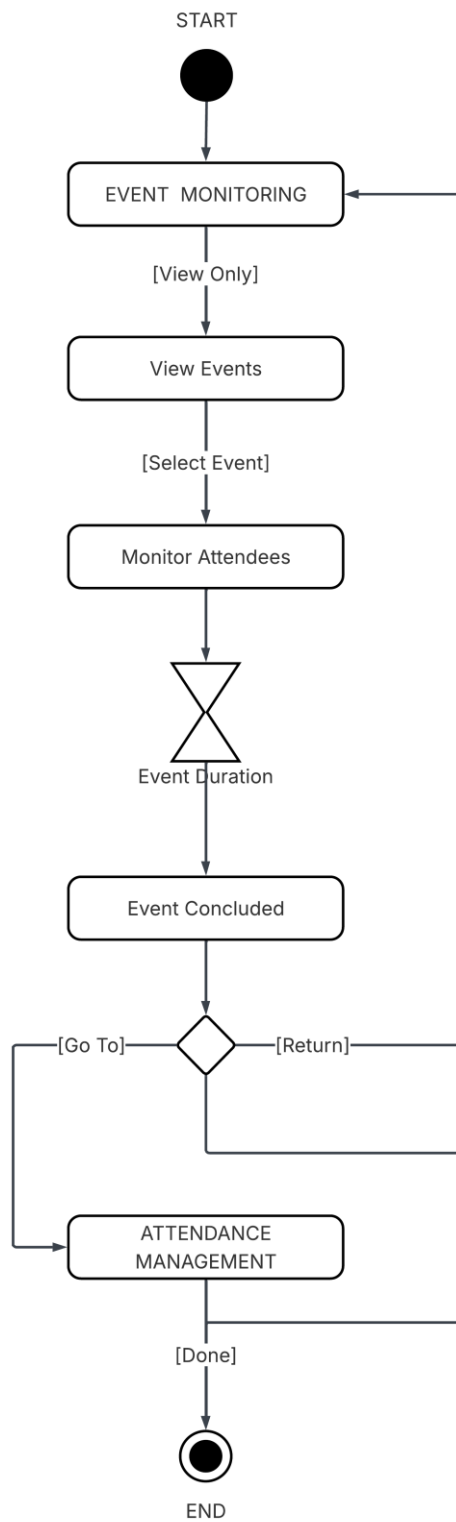


Figure 10. Activity Diagram of the OSA Module in Event Monitoring

The figure illustrates the attendance monitoring process, where the OSA coordinator views the list of events and selects one to monitor its attendees. The system tracks attendance for the duration of the event until it concludes. Afterward, the coordinator may return to monitor another event or proceed to Attendance Management to review and manage the recorded attendance data.

Figure 11 presents the activity diagram of the OSA module in attendance management.

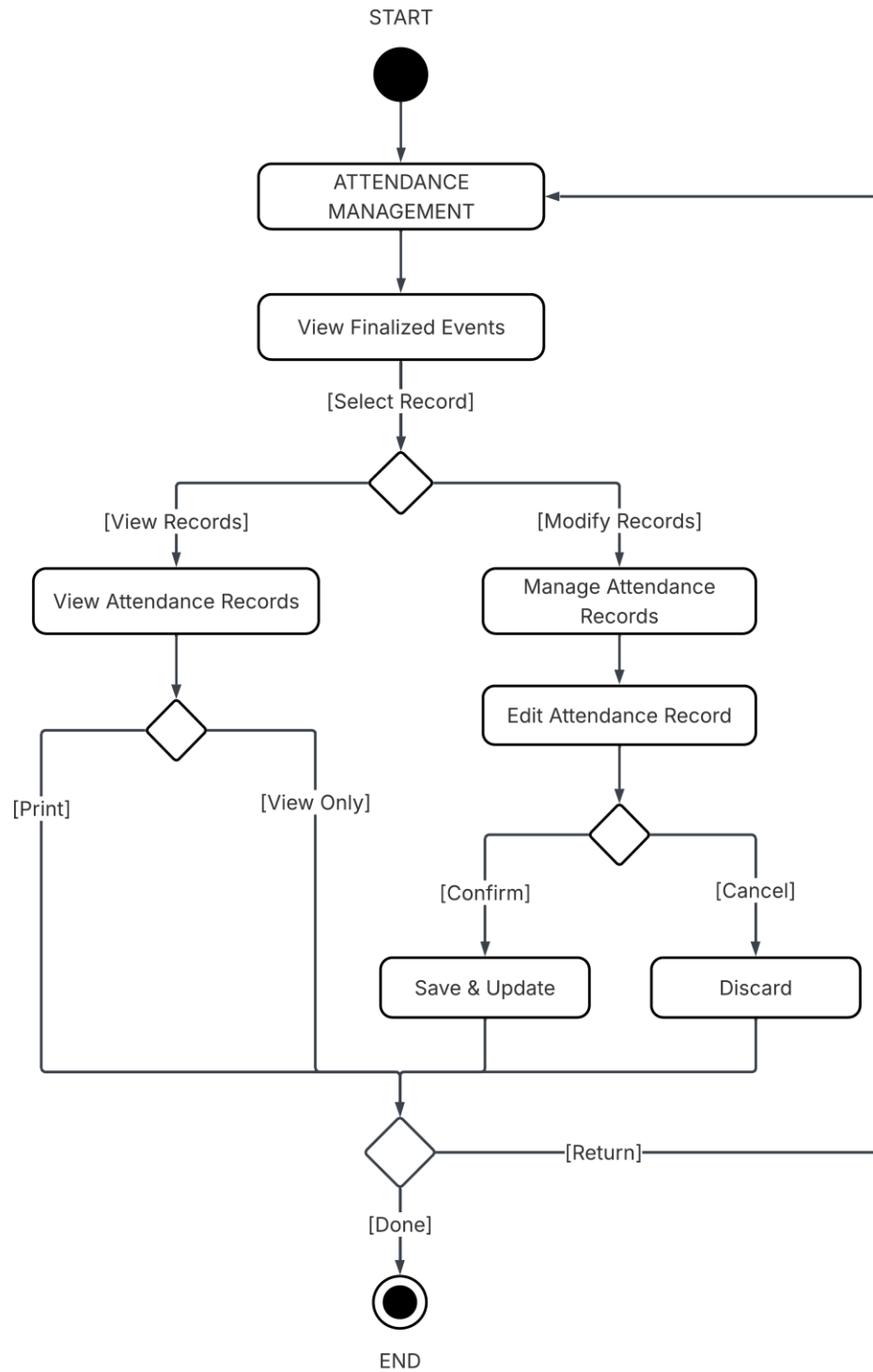


Figure 11. Activity Diagram of the OSA Module in Attendance Management

The figure illustrates the attendance management process, where the OSA coordinator views finalized events and selects one to access its attendance records. The coordinator may either view the records, with an option to print them, or modify them by editing an attendance record, which requires confirmation before the change is saved or discarded. After completing an operation, the coordinator may return to manage another event's attendance or exit the process.

Figure 12 presents the activity diagram of the OSA module in personal profile.

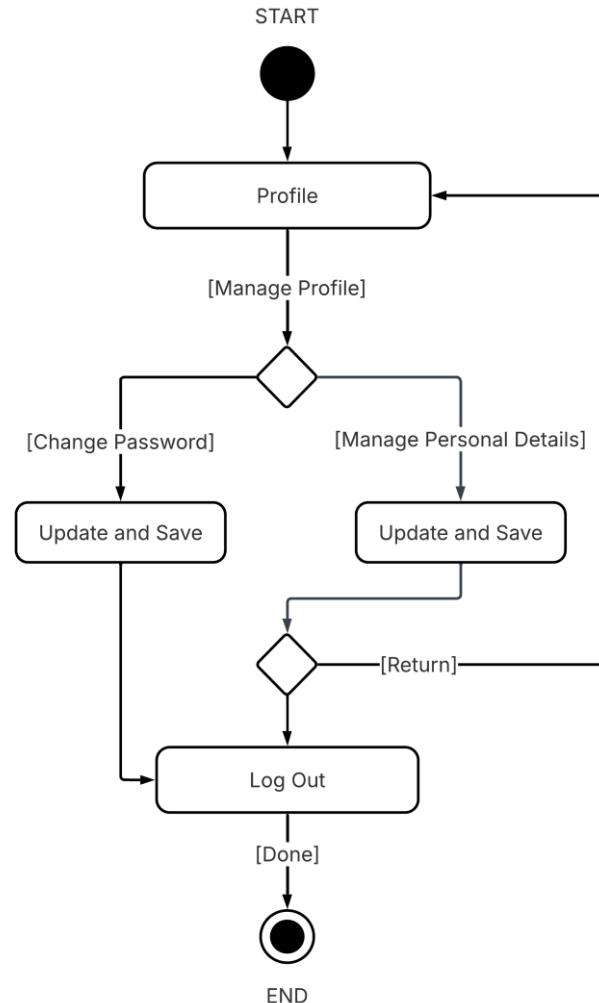


Figure 12. Activity Diagram of the OSA Module in Personal Profile

The figure illustrates the process beginning with the OSA staff accessing the Personal Profile module. Upon selecting Manage Profile, the staff is presented with two options: changing their password or updating their personal details. Both options lead to an Update and Save action that commits the changes to the system. Afterward, the staff may choose to return to the Profile module or log out to end the session.

Figure 13 presents the activity diagram of the student module in authentication and navigation flow.

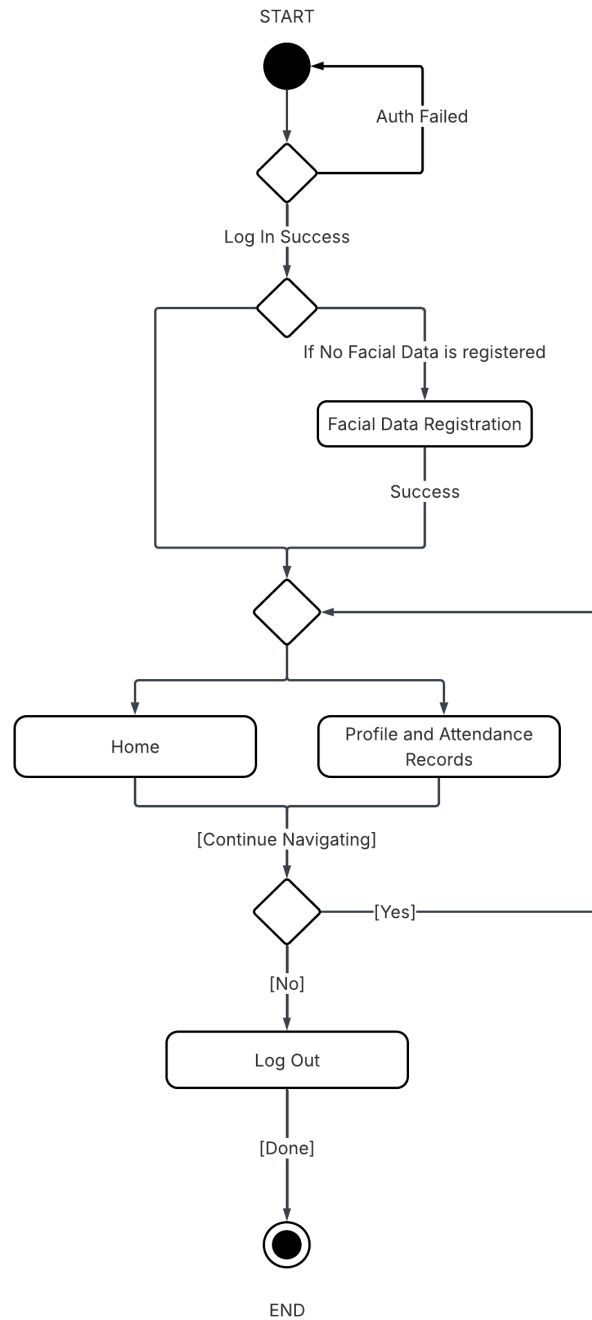


Figure 13. Activity Diagram of the Student Module in Authentication and Navigation Flow

The process begins when the student attempts to log in; if authentication fails, the system redirects the user back to the login screen to retry. Upon successful authentication, the system checks whether the student's facial data is already registered; if not, the user is directed to complete Facial Data Registration before proceeding. Once registration is confirmed, the user is granted access to two (2) pages: Home and Profile and Attendance Records. The student may continue navigating between pages as needed. Once the user chooses to end the session, the system proceeds to log out, terminating the process.

Figure 14 presents the activity diagram of the student module in facial data registration.

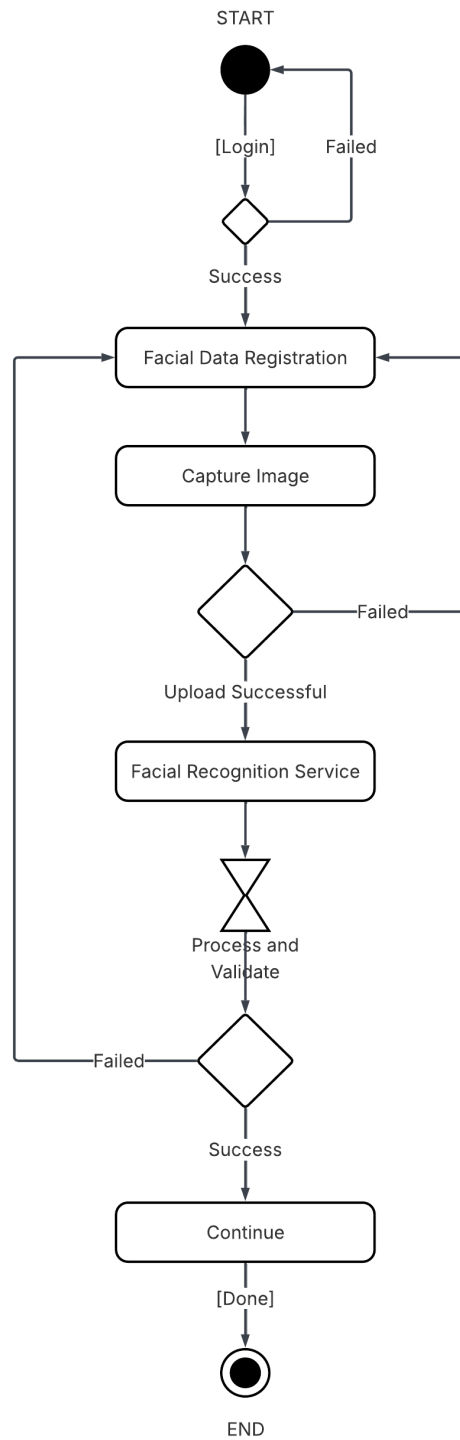


Figure 14. Activity Diagram of the Student Module in Facial Data Registration

The figure illustrates the process beginning with the student logging into the system using their school account. If it is the student's first time accessing the system, they are prompted to complete a one-time facial data registration. If the attempt fails on the students' side, the student is required to try again until the facial data is successfully captured. Once the facial data has been captured and uploaded, the system proceeds to validate it. If the validation is successful, the facial data registration is finalized. Otherwise, the student is prompted to repeat the registration process until validation is passed.

Figure 15 presents the activity diagram of the student module in event registration.

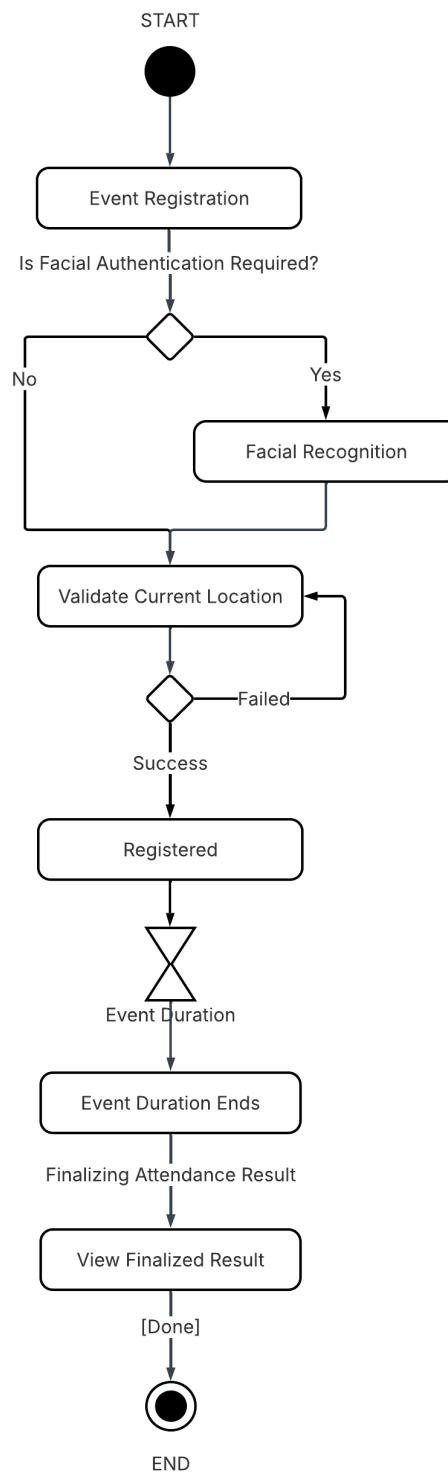


Figure 15. Activity Diagram of the Student Module in Event Registration

The figure illustrates the process of basic event registration. During registration, facial authentication may be required depending on the event configuration. If the authentication fails, the student is prompted to try again. Once facial authentication is successful, the system proceeds to validate the student's current location by checking whether they are within the designated geofenced area of the event. If the student's location is confirmed to be within the event boundaries, the system registers the student for the event. The mobile application then waits for any event status and duration changes.

Once the event concludes, location pinging stops and the system finalizes and presents the attendance records by calculating each student's attendance results.

Figure 16 presents the activity diagram of the Student Module in viewing profile and attendance records.

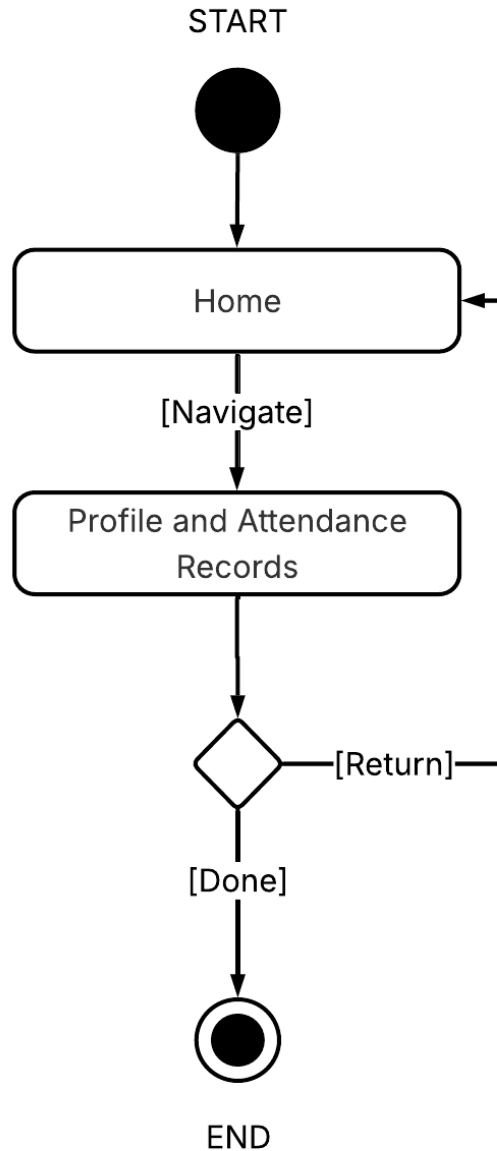


Figure 16. Activity Diagram of the Student Module in Viewing Profile and Attendance Records

The figure illustrates the process beginning with the student logging into the system using their school account. Upon successful login, the student is directed to the home where they can navigate to their profile and attendance records

module. From here, the student can view their personal profile information as well as their attendance records associated with concluded events. The attendance records display the student's attendance status for each event they were eligible for, indicating whether they were marked as present, idle, or absent. At the end of the process, the student may choose to log out and end the session or return to the view their personal profile.

Figure 17 presents the activity diagram of the Student Module in managing password and facial data inside settings.

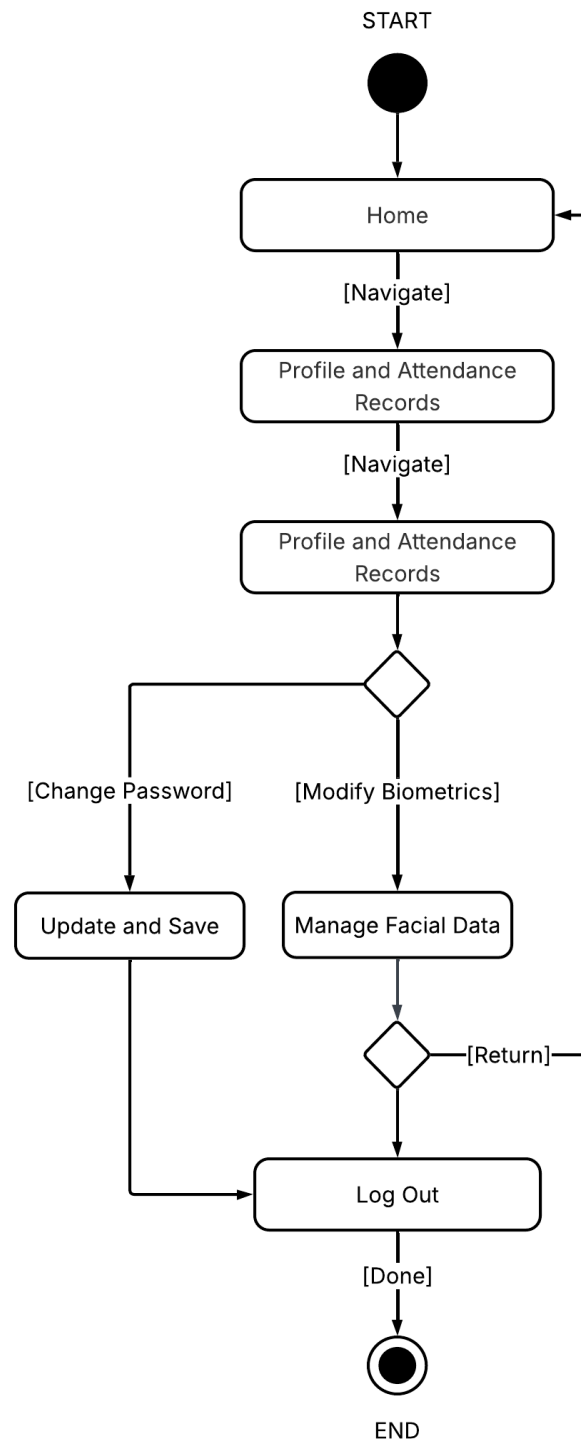


Figure 17. Activity Diagram of the Student Module in Managing Password and Facial Data inside Settings

The figure illustrates the student settings management process, where the student may either change their password or modify their biometric facial data. When changing their password, the student updates the necessary fields and saves the changes, after which they are automatically logged out and required to log back in. When modifying biometrics, the student manages their facial data and may choose to return and perform another operation or exit the process once done, without needing to log out.

Database Design

Table 1 shows the user collection, which holds the records of all registered users in the system.

Table 1. User collection (users)

Field Name	Data Type	Description
_Id	ObjectId	Auto-generated unique identifier for the user.
firstName	String	User's first name.
lastName	String	User's last name.
password	String	Encrypted user password.
contactNumber	String	Phone number.
email	String	Unique email address.
accountStatus	String(enum)	Activation status (default ACTIVE)
userType	String(enum)	Types of users (OSA, Student)
createdAt	Date	Timestamp indicating when this user record was added.
updatedAt	Date	Timestamp indicating the last modification to this user record.
updatedBy	String	Updater's ID.

This table captures core user information, including first name, last name, account password, and contact details such as phone number and email address, along with account status. It also includes audit fields including created at, updated at, and updated by to support traceability. The user collection serves as the central

store for personal information of all individuals in the system, including OSA staff and students.

Table 2 shows the student collection, which holds both basic student information and their associated biometric data.

Table 2. Student collection (students)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the student.
studentNumber	String	Unique institutional ID.
sectionName	String	Denormalized field referencing the section name on section collection.
courseName	String	Denormalized field referencing the course name on course collection.
clusterName	String	Denormalized field referencing the cluster name on course collection.
section	DBRef	Reference to the enrolled section.
currentSectionId	String	Denormalized field referencing the section collection pointing to the section.sectionId.
user	DBRef	Reference to the user collection.
userId	String	Denormalized field referencing the user collection pointing to the user.userId.
facialData	DBRef	Reference to the biometric facial data.

This table covers student-specific records, including user references, biometric data, and section assignments for enrollment management. It serves as the primary store for student data within the system, consolidating personal and biometric information to support event registration, attendance recording, and monitoring.

Table 3 shows the biometric data collection, which stores facial recognition data used to verify student identity.

Table 3. Biometric Data collection (biometric_data)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the biometric record.
faceEncoding	Array<Float>	Fixed-size vector (128 elements) of facial embeddings for recognition.
studentNumber	String	Denormalized field referencing the student number on student collection.
createdAt	Date	Timestamp indicating when this biometric data record was created.
lastUpdated	Date	Timestamp indicating the last modification to this biometric data record.
biometricStatus	String(enum)	Status of biometric enrollment (PENDING, ACTIVE, INACTIVE). Defaults to PENDING.

This table contains biometric enrollment details referenced by the student collection, including facial feature vectors and status tracking for authentication management. It serves as the secure store for student biometric templates, enabling facial recognition during event registrations when it is required.

Table 4 shows the academic year collection, which holds all fields related to academic year records.

Table 4. Academic Year collection (academic_year)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the academic year document.
academicYearName	String	Name of a specific academic year.
currentSemester	String	Name of the current active semester.
firstSemesterStart	Date	Starting date of the first semester.
firstSemesterEnd	Date	Ending date of the first semester.
secondSemesterStart	Date	Starting date of the second semester.
secondSemesterEnd	Date	Ending date of the second semester.
isActive	Boolean	Status flag of the specific academic year.
createdAt	Date	Timestamp indicating when this academic year record was created.
updatedAt	Date	Timestamp indicating the last modification to this academic year record.

This table represents the academic year, structured around a two-semester calendar. Each academic year defines distinct first and second semesters, each with corresponding start and end dates, and may be associated with section records.

Table 5 shows the section collection. The section collection contains all academic enrollment sections.

Table 5. Section collection (section)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the section.
sectionName	String	Full section identifier.
yearLevel	Integer	Field on what year level the section should belong to.
semester	Integer	Field on what semester the section should belong.
courses	DBRef	Reference to the parent document, course.
isActive	Boolean	Status flag of the specific section.
createdAt	Date	Timestamp indicating when this section record was created.
updatedAt	Date	Timestamp indicating the last modification to this section record.

This table captures specific details, including section name, year level, semester, and course associations, along with audit timestamps for academic enrollment management. It stores granular student groupings within the system, enabling targeted event eligibility, targeted attendance tracking, and filtered student account management.

Table 6 shows the course collection. The course collection contains all academic enrollment courses.

Table 6. Course collection (course)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the course.
courseName	String	Unique course code/name.
cluster	DBRef	Reference to the parent cluster document.
createdAt	Date	Timestamp indicating when this course record was created.
updatedAt	Date	Timestamp indicating the last modification to this course record.

This table contains academic course details, including program names and cluster associations, along with audit timestamps for curriculum management. It stores the mid-level educational entities within the system, linking clusters to sections and enabling targeted event invitations for course-specific student groups.

Table 7 shows the cluster collection. The cluster collection contains all academic enrollment clusters.

Table 7. Cluster collection (cluster)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the cluster.
clusterName	String	Unique name for the academic cluster.
createdAt	Date	Timestamp indicating when this cluster record was created.
updatedAt	Date	Timestamp indicating the last modification to this cluster record.

This table defines groupings for academic programs, including cluster names and audit timestamps for organizational management. It stores top-level course categorizations within the system, supporting broad eligibility rules for events and attendance tracking and filtered attendance monitoring.

Table 8 shows the attendance records collection. It is used to record the attendance of the students at every event.

Table 8. Attendance Records collection (attendance_records)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the record.
student	DBRef	Referencing the student document.
studentId	String	Denormalized field referencing the student number on student document.
studentUserId	String	Denormalized field referencing the user id on user document.
event	DBRef	Referencing the event document.
location	DBRef	Referencing the location document.
eventLocationId	DBRef	Denormalized field referencing the location id on location document.
academicYear	DBRef	Referencing the academic year document.
academicYearId	String	Denormalized field referencing the academic year id on academic year document.
academicYearName	String	Denormalized field referencing the academic year name on academic year document.
semester	Integer	Denormalized field for identifying current semester number.
semesterName	String	Denormalized field for identifying current semester name.
timeIn	Date	Time-in timestamp.
timeOut	Date	Time-out timestamp.
reason	String	Explanation for students' attendance status.
attendanceStatus	String(enum)	Final attendance status of the student (PRESENT, ABSENT, REGISTERED, IDLE, LATE, EXCUSED)
attendancePingLogs	Array<Object>	List of attendance tracking pings. Defaults empty.
updatedByUserId	String	Updater's user ID.
createdAt	Date	Timestamp indicating when this attendance record was added.
updatedAt	Date	Timestamp indicating the last modification to this attendance record.

This table captures student-event attendance details, including registration timestamps, attendance status, location ping logs, and attendance remarks, along with audit timestamps for record management. It stores verified participation data

within the system, supporting finalization calculations and reporting for academic compliance.

Table 9 shows the location collection. It is used to track the location of the students' device.

Table 9. Location collection (location)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the location.
locationName	String	Descriptive name of location.
description	String	An optional description for location.
locationEnvironment	String	Environment type of location (INDOOR or OUTDOOR).
locationPurpose	String	Use of the specific location (REGISTRATION_AREA or VENUE).
geometryType	String	Geometry format.
geometry	GeoJsonPolygon	Spatial boundaries for geofencing (2dsphere indexed).
createdAt	Date	Timestamp indicating when this location record was created.
updatedAt	Date	Timestamp indicating the last modification to this location record.

This table holds the location and geospatial details that are used for events. This collection stores fields including the location name, location types, and polygon boundaries, along with audit timestamps for traceability. Each location is classified by its purpose, whether it serves as a registration gateway or a primary event venue, and by its environment type to inform signal accuracy expectations during event setup. It stores physical or virtual sites within the system, supporting geofenced attendance verification to confirm students are within designated event boundaries.

Table 10 shows the event collection. The event collection contains information about the event sessions.

Table 10. Event collection (event)

Field Name	Data Type	Description
_id	ObjectId	Auto-generated unique identifier for the event.
eventName	String	Descriptive name of event.
registrationLocation	DBRef	Referencing to the location document. The specific area where students must physically be present to register for the event.
registrationLocationId	String	Denormalized field referencing location id.
registrationLocationName	String	Denormalized field of location name for the name of registration location.
venueLocation	DBRef	Referencing to the location document. The actual space where the event takes place.
venueLocationId	String	Denormalized field referencing location id.
venueLocationName	String	Denormalized field of location name for the name of venue location.
description	String	Detailed event overview.
eligibleStudents	Object	Eligibility criteria (yearLevel/sections/courses/clusters).
registrationDateTime	Date	Registration opening time.
startingDateTime	Date	Official start of the event.
endingDateTime	Date	Event end time.
eventStatus	String(enum)	Current state of the event (UPCOMING, REGISTRATION, ONGOING, CONCLUDED, CANCELLED, FINALIZED)
facialVerificationEnabled	Boolean	Flag for enabling/disabling facial biometric verification during event registration.
attendanceLocationMonitoringEnabled	Boolean	Flag that enabled continuous location monitoring during the event.
strictLocationValidation	Boolean	Enables strict event location registration validation.
createdByUserId	String	Identifier of the user who created this event.
createdAt	Date	Timestamp indicating when this event record was created.
updatedAt	Date	Timestamp indicating the last modification to this event's record.

This table captures event scheduling and eligibility details, including timings, locations, and status tracking, along with audit timestamps for session

management. It also supports a dual-location model which is a technique to verify two (2) times the presence of the student within the two (2) locations, enabling precise attendance tracking. It stores academic and administrative events within the system, facilitating registration, geofenced attendance, and post-event record finalization for students.

Implementation Phase

Based on the output, the development of RCians Attendease involved a range of tools and technologies across its interconnected system components.

The developers designed the overall system architecture, including how its components interact with one another. A custom user management module handling user authentication and role-based access control for two defined roles (OSA and Student), along with other required user fields, was developed in Java with Spring Boot and integrated within the same backend alongside the system's core features: attendance tracking and monitoring, and location, academic data, attendance, and event management. The web application, used by OSA to manage users, locations, academic data, and manage and monitor events, and the mobile application, used by students to view and register attendance, were both integrated with the backend, with all required modules for each component fully implemented. A dedicated facial recognition microservice was also built using Python and the OpenCV library to support image processing for facial registration and verification, operating as an independent component that communicates with the backend via REST API. Git was used for version control throughout

development, while GitHub served as the remote repository for source code management and collaboration among developers. Figma was used for designing the system's user interface and prototyping screens prior to implementation. Visual Studio Code and IntelliJ IDEA served as the primary code editors, with Visual Studio Code used mainly for the web and mobile application codebases, and IntelliJ IDEA used for developing the Java Spring Boot backend.

The completed system was hosted on Amazon Web Services and deployed manually through the AWS CLI by transferring the application's executable JAR file directly to an EC2 instance.

Integration and Testing Phase

Alpha testing was conducted during several development phases with developers and the project adviser to identify and resolve issues, ensuring the system was stable and ready to proceed to Beta testing which was conducted to assess the system's behavior in real-world conditions, where RCians Attendease was deployed on a stable hosting environment on AWS. A select group of Rogationist College - College Department students observed the system across various scenarios and provided feedback on its performance, usability, and overall behavior. Throughout this phase, system testing was conducted in accordance with the ISO 9126 quality standard, focusing on selected quality attributes relevant to the system's evaluation: functionality, usability, reliability, and efficiency.

Deployment of System Phase

Once the system had passed the beta testing phase, RCians Attendease was deemed ready for release to its intended end users. The developers also ensured that the system remained redeployable and portable, allowing it to be reliably reproduced across different environments. Following deployment, user suggestions, comments, and feedback were collected and noted for future reference and further development.

Maintenance Phase

As the final phase of the waterfall model, this stage will ensure that the system continues to operate as intended. Any bugs or errors on that surface will be identified and resolved, while improvements and new features will be incorporated into the system as needed.

Operation and Testing Procedure

Table 11 presents the criteria and testing procedures used by the developers to evaluate the functionality and overall quality of the system across both the web and mobile applications. Each criterion is based on the ISO 9126 quality attributes and outlines the specific steps taken to verify that the system meets the expected standards for each sub-attribute.

Table 11. Criterion's Testing Procedure

Criterion/Sub-Criteria	Testing Procedures
1. Functionality a. Suitability b. Accuracy c. Security	• Open the web and mobile application. • Access the web and mobile application using computer and mobile phones. • Ensure that the buttons are properly working. • Ensure that the system will display the desired result. • Ensure that the system will display if the student attended an event. • Check if the student logs to an event corresponding with their activity during the event. • Check if the system can verify if the user ID is unique and valid. • Check if the given username and password from the student affairs are correct and can be used to the system. • Secure the data of each module that has the function of altering the data.
2. Reliability	• Run the system using both the web and mobile application on a desktop and smartphone. • Restart the app while doing a task to see if it can continue properly and save your progress.
3. Usability a. Understandability b. Learnability	• Access the different modules to each client, student affairs and students. • Analyze the presented information of the modules that are given by the system. • Analyze if the user modules have proper naming for each operation.

Table 12. Continuation of Criterion's Testing Procedure

Criterion/Sub-Criteria	Testing Procedures
3. Usability	
c. Operability	• Access the different modules to each client, student affairs and students.
d. Attractiveness	• Analyze the presented information of the modules that are given by the system. • Analyze if the user modules have proper naming for each operation. • Observe if a user can easily understand how to use the system without experience. • Observe if the system meets the requirements of all user modules • Observe if the user interface of desktop, web, and mobile applications is attractive to the end users.
4. Efficiency	
a. Time Behavior	• Observe if the system completes processes within a minimal time to ensure operational efficiency.
b. Resource Utilization	• Observe if the system responds appropriately and in real-time during events. • All resources and user modules should be functional and utilized during the testing process. • Observe if the system can handle huge amounts of data. • Observe if the system shall be capable of processing a high volume of data during normal transactional operations.
5. Maintainability	
a. Changeability	• Check the system's design and modules to make sure each part is separate and easy to update. • Check if the change can be implemented without affecting other parts of the system.

Table 13. Continuation of Criterion's Testing Procedure

Criterion/Sub-Criteria	Testing Procedures
5. Maintainability	
b. Testability	• Test individual modules to verify that each component works as expected. • Check if the system's logs and error messages provide enough information to troubleshoot issues effectively.
6. Portability	
a. Adaptability	• Evaluate system's architecture, ensuring it can operate across different operating systems and hardware platforms without requiring significant modifications. • Verify that the system uses platform-agnostic technologies and avoids dependencies on specific hardware or software configurations.
b. Installability	• Assess the installation process to ensure it is straightforward, well documented and compatible with target environments. • Check for automated installation scripts or tools that minimize manual configuration and reduce risk of errors during setups. • Verify the availability of installers for each platform.
c. Replaceability	• Ensure that system components or modules can be replaced with alternative implementations without impacting overall functionality. • Ensure the system adheres to standard interfaces and modular design principles to facilitate seamless replacement of components. • Ensure the system replaceability of a new version when upgrading.

Evaluation Procedure

The following procedures are used to evaluate the RCians Attendease.

Preliminary Evaluation

The developers, together with their adviser, conducted a preliminary evaluation of the facial recognition and geofencing-based attendance monitoring system at Rogationist College to verify that all modules were functioning as intended.

Final Evaluation

The final evaluation was conducted by the Coordinator of the Office of Student Affairs and a sample of 10 percent of the total college student population at Rogationist College. Evaluators assessed the system using the ISO 9126 standard. The developers took note of the feedback raised by the evaluators and incorporated it into the study documentation.

Evaluation Instrument

The International Organization for Standardization (ISO) 9126 model was used to assess the quality of the geofencing-based attendance monitoring system. According to Umar (2019), ISO 9126 is one of the most comprehensive and widely adopted models for measuring software quality. As a general-purpose model, it can be applied across different types of software. The model encompasses six (6) key quality attributes: functionality, reliability, usability, efficiency, maintainability, and portability.

Functionality

Functionality refers to the ability of the geofencing-based attendance monitoring system to fulfill its assigned tasks and meet its intended purpose. It encompasses four (4) sub-attributes: suitability, accurateness, interoperability, and security. For this evaluation, three (3) of these sub-attributes were considered: suitability, accurateness, and security. Interoperability was excluded, as the system was not designed to integrate with any external structure.

Reliability

Reliability refers to the system's ability to perform consistently under specified conditions. It encompasses three (3) sub-attributes: maturity, which measures how rarely the system produces failures under normal use; fault tolerance, which refers to the system's ability to continue operating when errors or unexpected inputs occur; and recoverability, which describes how quickly and effectively the system can restore normal operation after a failure. However, these sub-attributes were not individually assessed in this evaluation, as reliability was measured holistically based on the system's overall ability to perform consistently under normal operating conditions, without necessitating a breakdown into specific sub-criteria.

Usability

Usability refers to the ease with which users can understand and operate the system. It encompasses four (4) sub-attributes: understandability,

which measures how easily users can grasp the system's purpose and functions; learnability, which refers to how quickly users can learn to operate the system effectively; operability, which describes how smoothly and efficiently users can interact with the system; and attractiveness, which reflects the visual appeal of the system's interface. All four (4) sub-attributes were included in the evaluation to ensure that the system met the necessary criteria for an intuitive and pleasant user experience.

Efficiency

Efficiency refers to how effectively RCians Attendease performs in relation to time behavior and resource utilization. Both sub-attributes were included in the evaluation: time behavior, which measures how quickly and responsively the system performs under real-time conditions, and resource utilization, which assesses how optimally the system makes use of available resources including CPU, RAM, and network bandwidth on both the server and client sides. Both sub-attributes were incorporated given the system's requirement to respond in real time while maintaining efficient use of its resources.

Maintainability

Maintainability refers to how easily a system can be modified, updated, or repaired over time, ensuring it remains functional as new requirements or issues arise. It encompasses four (4) sub-attributes: analyzability, which measures how easily faults or causes of failure can be identified; changeability, which refers to

how readily the system can accommodate new features or improvements; stability, which describes the system's ability to avoid unexpected behavior following modifications; and testability, which reflects how effectively the system can be validated after changes are made. For this evaluation, maintainability was assessed through changeability and testability, as these sub-attributes were most relevant to ensuring the system could be extended and verified with confidence. While analyzability and stability were excluded, as the system was still in its initial deployment stage and hadn't undergone any maintenance which these attributes assessed.

Portability

Portability refers to the system's ability to adapt and function across different environments. It encompasses four (4) sub-attributes: adaptability, which measures how well the system can operate across varying environments without requiring modifications; installability, which refers to how easily the system can be installed in a given environment; co-existence, which describes the system's ability to operate alongside other systems in a shared environment; and replaceability, which reflects how readily the system can be substituted by another system with comparable functionality in the same environment. For this evaluation, adaptability, installability, and replaceability were included, as the system was required to be deployable across different environments, function without environment-specific modifications, and be replaceable when necessary. Co-

existence was excluded, as the system was not designed to operate alongside any separate external systems.

Statistical Treatment of Data

The system will be evaluated using the International Standard Organization 9126 (ISO 9126) that is also used in the study of Uy et al. (2022).

Table 14 represents the scoring system

Table 14. Evaluation Scoring System Table

Numerical Scale	Description
3	Excellent
2	Good
1	Fair
0	Poor

The table is separated into four (4) independent measurements. Zero (0) is the lowest and three (3) as the highest.

Table 15 shows the priority value of International Standard Organization 9126 (ISO 9126) that has a descriptive meaning for the assessment score.

Table 15. Prioritization of Characteristics Table

Characteristics	Priority (Pr)
Functionality	6
Reliability	2
Usability	3.5
Efficiency	1
Maintainability	5
Portability	3.5

The table displays six (6) characteristics outlined in the International Standard Organization 9126 (ISO 9126). Each characteristic is assigned priority

values, where functionality is rated at six (6), reliability at two (2), usability at three point five (3.5), maintainability at five (5), and portability at three point five (3.5).

The following formulas were used to find out the quality factor:

$$TQVS = \sum FQVC$$

$$FQVC = QVC * Pr$$

$$QF = TQVS / ITQVS$$

Wherein:

QVC = average of each characteristic

The total average score of the system (TQVS) is found by adding the scores of each feature (FQVC). To find out how good the system is overall, the researchers divide that total score by the best possible score it could get (ITQVS). This result is called the quality factor (QF), which shows how high or low the system's quality is. The best possible score (ITQVS) is based on the idea that all features were rated a perfect three (3) by the evaluators.

Table 16 shows the range of the mean value table.

Table 16. Range of Mean Value

Range of Mean Value	Description	Interpretation
0.76 – 1.00	Excellent	The system had met all the specific requirements, with functionality, reliability, usability, efficiency, maintainability, and portability.
0.51 – 0.75	Good	The system had met most of the specific requirements, with functionality, reliability, usability, efficiency, maintainability, and portability.
0.26 – 0.50	Fair	The system had met some of the specific requirements, with functionality, reliability, usability, efficiency, maintainability, and portability.
0.00 – 0.25	Poor	The system had met a few of the specific requirements, with functionality, reliability, usability, efficiency, maintainability, and portability.

For the application to be approved, the system needs to achieve a mean value of at least 0.51.

Chapter 4

RESULTS AND DISCUSSION

This chapter contains the project description, project structure, and project evaluation.

Project Description

RCians Attendease served as a web and mobile application designed to automate event attendance tracking for the College Department of Rogationist College. Developed with two interconnected modules, namely the Office for Student Affairs (OSA) module and the Student module, the system reflected the structure outlined in the activity diagram of the previous chapter.

Figure 18 presents the logo of the RCians Attendease.



Figure 18. Logo of the RCians Attendease

The figure displayed the official logo of the RCians Attendease. The system streamlines attendance management by replacing the traditional manual process with an automated approach, confirming student presence through device-based geolocation and facial recognition verification.

Project Structure

The OSA module served as a web-based application that enabled the OSA Coordinator to manage events, locations, and user accounts, as well as monitor and manage attendance records. The Student module, on the other hand, served as a mobile application that verified students' identity through facial recognition upon login. It utilized the device's GPS for geofenced event location validation, and allowed students to record their attendance and view available events and their details.

Figure 19 displays the login interface of the OSA module.

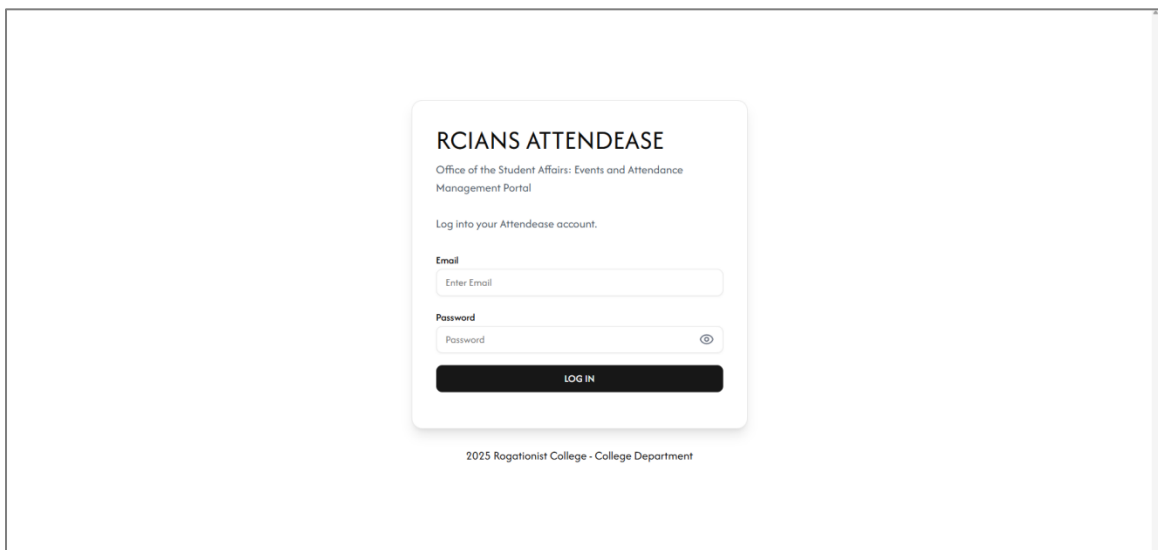
The image shows a web-based login interface for 'RCIANS ATTENDEASE'. The title 'RCIANS ATTENDEASE' is at the top, followed by the subtitle 'Office of the Student Affairs: Events and Attendance Management Portal'. Below this is the instruction 'Log into your Attendease account.' The form contains two input fields: 'Email' with a placeholder 'Enter Email' and 'Password' with a placeholder 'Password' and a toggle icon for visibility. A black 'LOG IN' button is positioned below the password field. At the bottom of the form, it says '© 2025 Rogationist College - College Department'.

Figure 19. Login Interface of the OSA module

The figure displays the OSA login form, which requires an email address and a password to access the system. The developers provided a default email address and password for the OSA coordinator, ensuring a secure and controlled login process. Additionally, the form includes a login button to proceed, and a show password option to make the password visible while typing.

Figure 20 displays the academic management interface for the OSA Module.

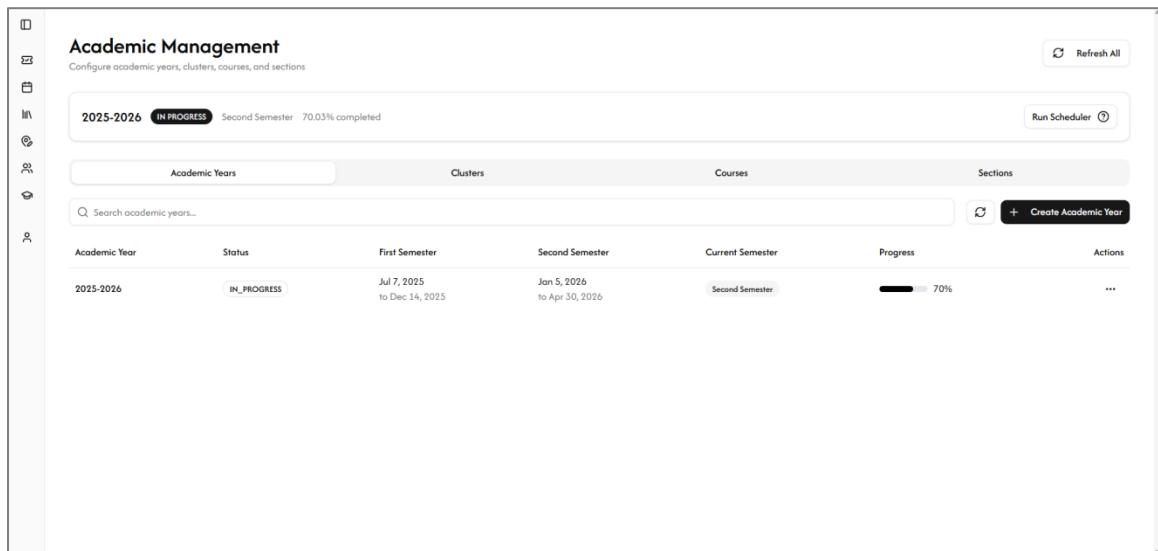


Figure 20. Academic Management Interface

The figure displays the Academic Management module, which is organized into four (4) sections. The Academic Years section is used to manage academic year records including their semester schedules and current progress. The Clusters section is used to manage cluster groupings within the college department. The Courses section is used to manage the courses offered under each cluster. The Sections section is used to manage the specific class sections assigned to each course.

Figure 21 displays the location management interface for the OSA Module.

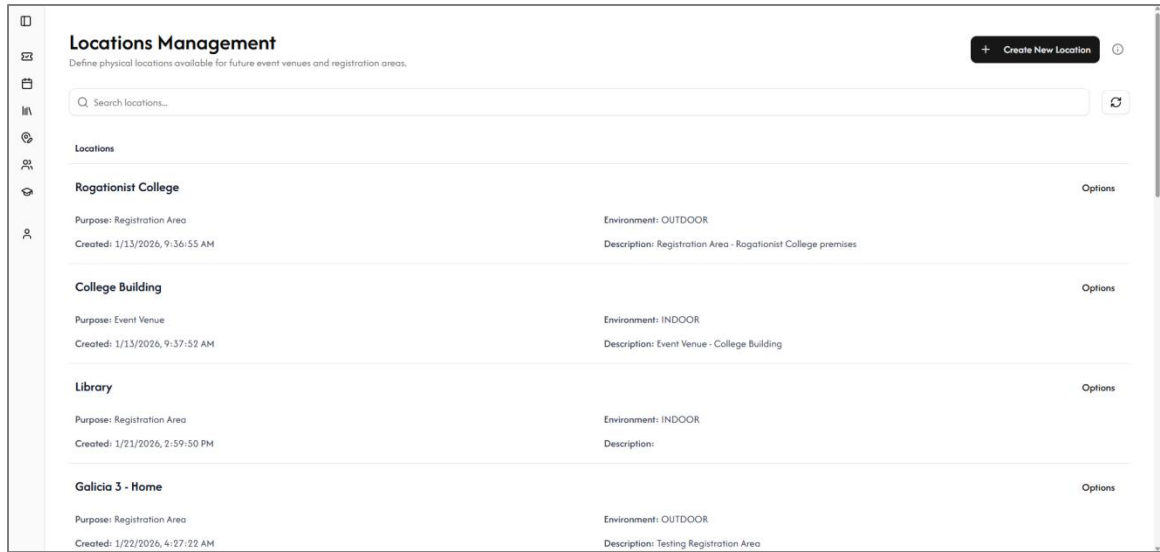


Figure 21. Location Management Interface

The figure displays the Location Management module, which allowed the OSA Coordinator to manage all venue records within the system. It enables the coordinator to search, create, edit, and delete locations used as registration areas or event venues throughout the system.

Figure 22 displays the user management interface for the OSA Module.

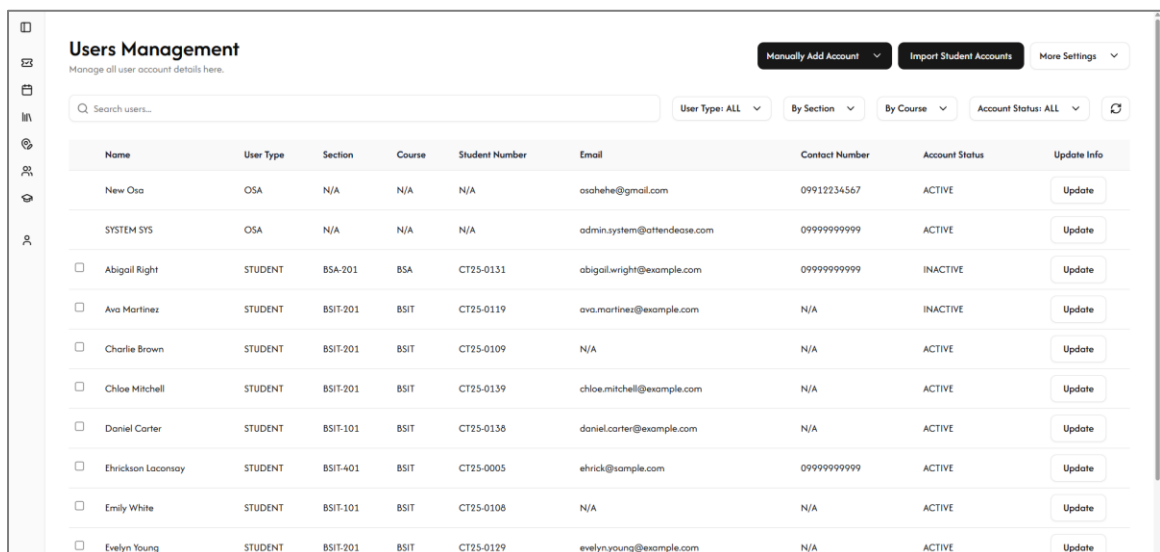


Figure 22. Users Management Interface

The figure displays the User Management module, which allowed the OSA Coordinator to manage all registered user accounts within the system. It enables the coordinator to search, filter, and view users by their cluster, section, and course. The module also supported adding individual accounts manually, importing multiple student accounts through a CSV file, and performing bulk account operations such as deleting all accounts or deleting accounts by section.

Figure 23 displays the event management interface for the OSA Module.

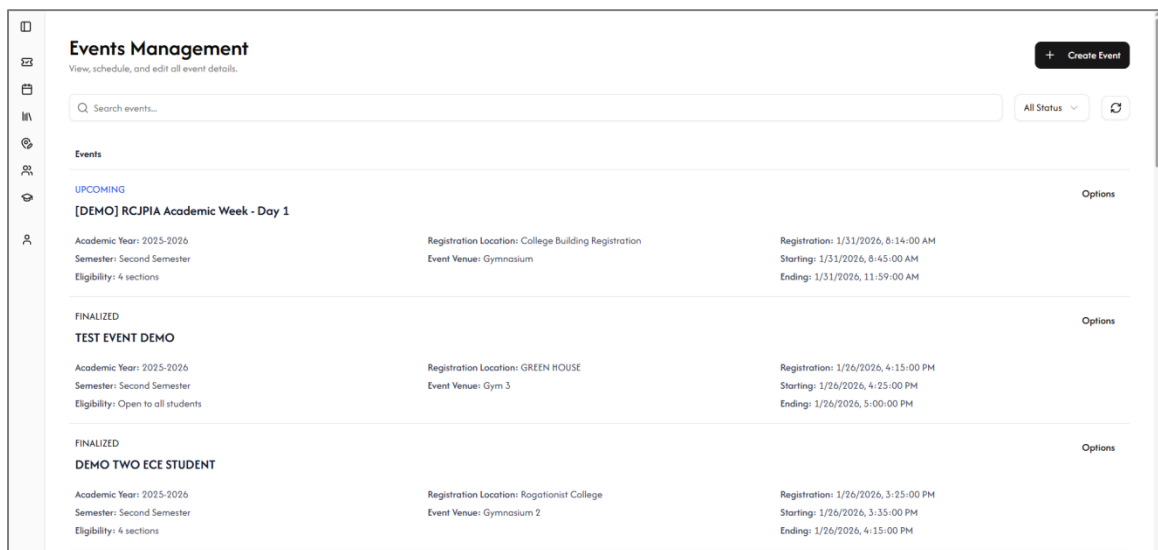


Figure 23. Event Management Interface

The figure displays the Event Management module, which allowed the OSA Coordinator to manage all events within the system. It enables the coordinator to create, search, filter, edit, and delete events, as well as monitor each event's status throughout its lifecycle from upcoming to finalized.

Figure 24 displays the event monitoring interface for the OSA module.

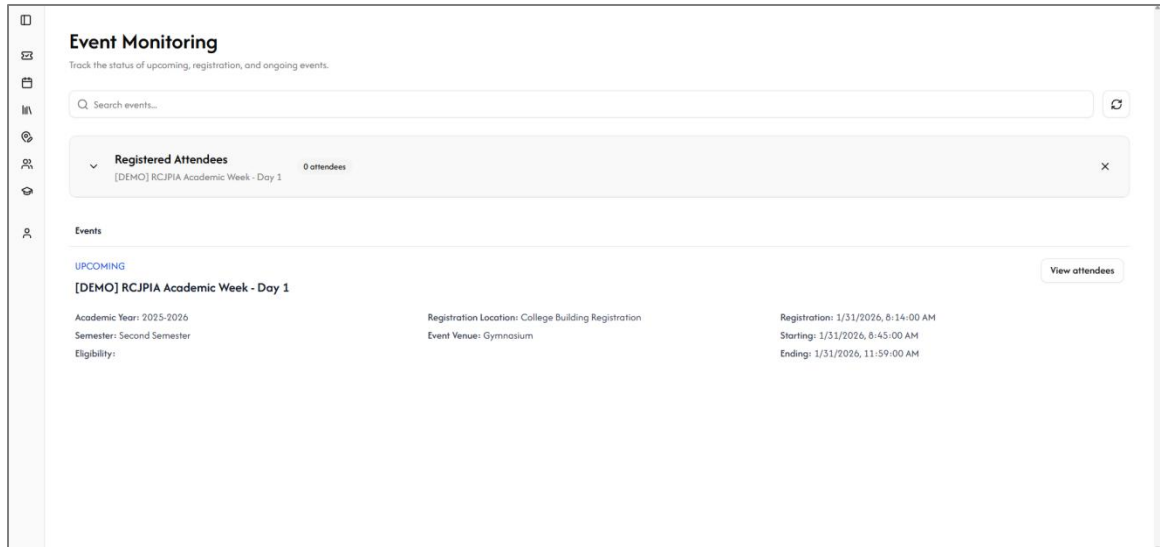


Figure 24. Event Monitoring Interface

The figure displays the Event Monitoring module, which allowed the OSA Coordinator to monitor active events limited to those with upcoming, registration, and ongoing statuses. It enables the coordinator to search for specific events and view the list of registered attendees for each monitored event.

Figure 25 displays the attendance management interface for the OSA module.

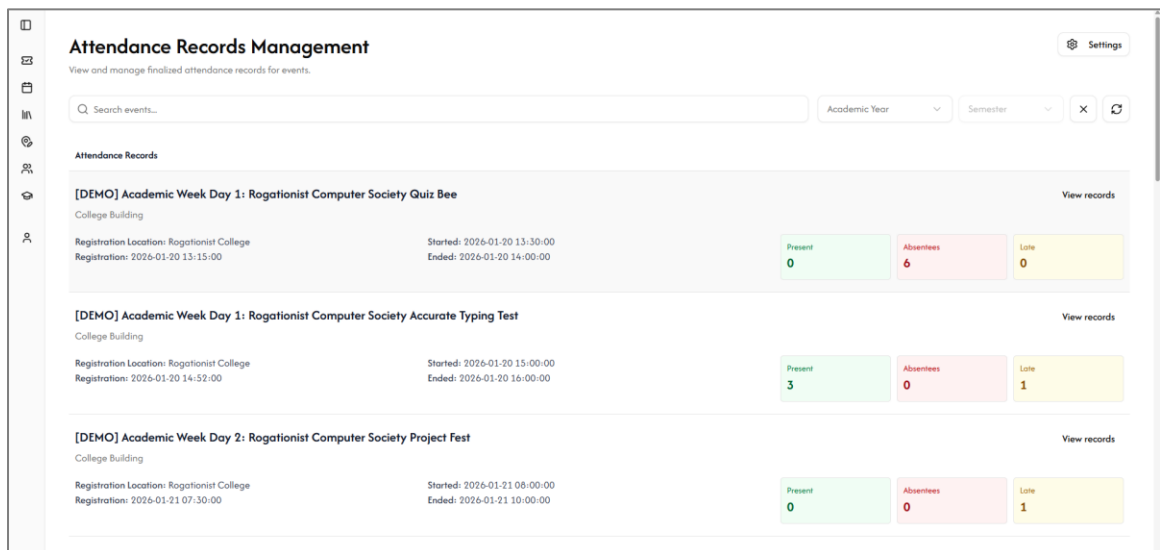


Figure 25. Attendance Management Interface

The figure displays the Attendance Management module, which allowed the OSA Coordinator to manage and review attendance records across all concluded events. It enables the coordinator to search, filter records by academic year and semester, and view the detailed attendance breakdown of each event including the number of students marked as present, absent, and late.

Figure 26 displays the personal profile interface for the OSA Module.

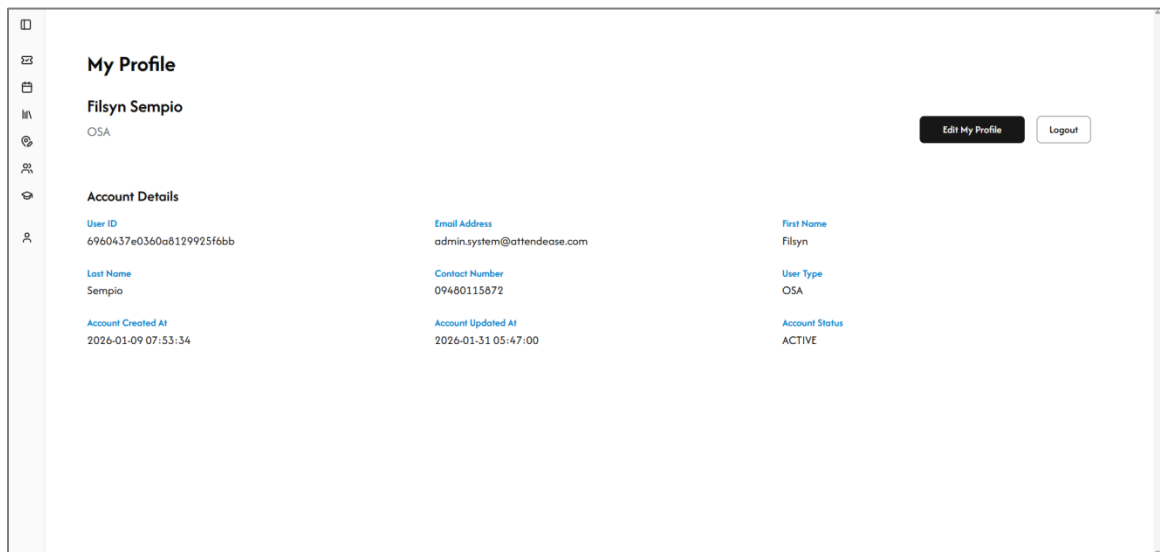
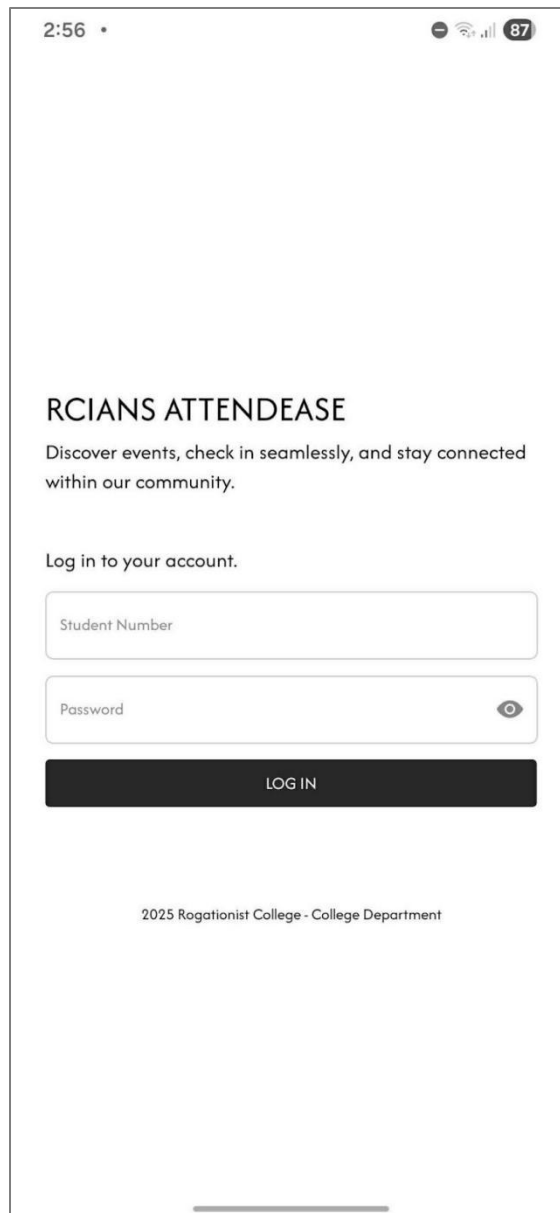


Figure 26. Personal Profile Interface

The figure displays the Personal Profile module, which allowed the OSA Coordinator to view and manage their personal account information. It also provide options to edit their profile details and log out of the system.

Figure 27 displays the interface of the login interface for the Student module.



The image shows a mobile application login screen. At the top, the status bar displays the time 2:56, signal strength, and battery level at 87%. The main heading is "RCIANS ATTENDEASE" in bold, followed by the subtitle "Discover events, check in seamlessly, and stay connected within our community." Below this, the text "Log in to your account." is displayed. There are two input fields: "Student Number" and "Password". The "Password" field has a toggle icon (an eye) to the right. A black "LOG IN" button is positioned below the input fields. At the bottom, the text "2025 Rogationist College - College Department" is visible.

Figure 27. Login Interface of the Student Module

The figure displays the Login interface of the Student module, which served as the initial screen upon opening the mobile application. It allows students to authenticate themselves into the system using their institutional student number and assigned initial password.

Figure 28 displays the interface of the facial data registration for the Student module.

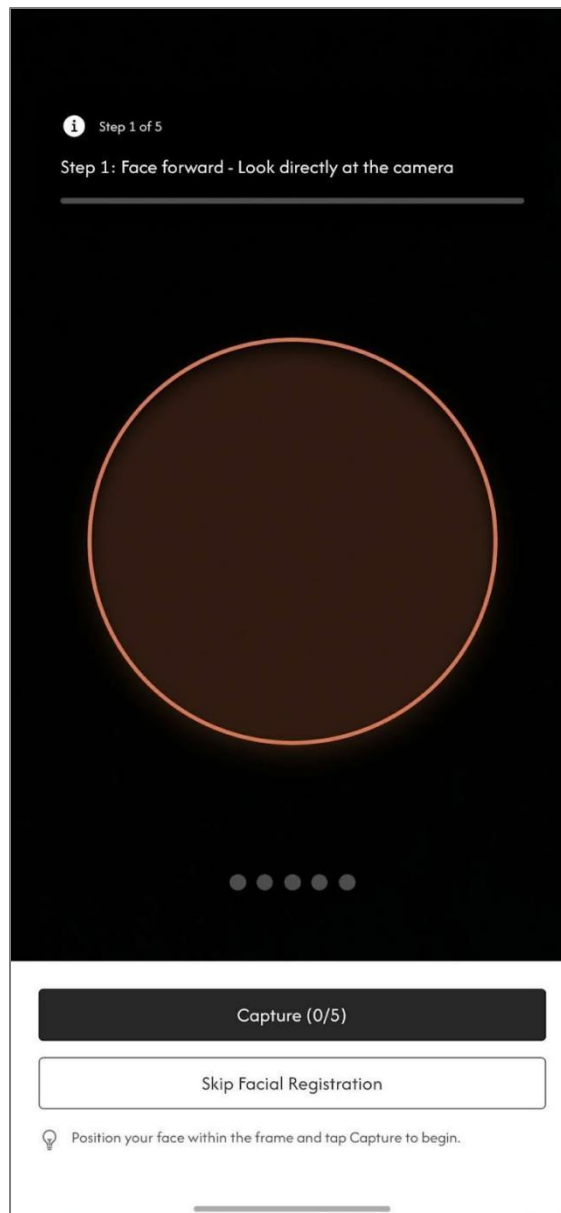


Figure 28. Facial Data Registration Interface

The figure displays the Facial Data Registration interface, which allowed students to register their facial data using the device's front camera. It guides students through the capture process and provide an option to skip facial registration if needed.

Figure 29 displays the interface of the event registration for the Student module.



Figure 29. Event Registration Interface

The figure displays the Event Details and Registration interface, which allowed students to view comprehensive information about a specific event including its schedule, location, and registration requirements. It also enables students to register for the event, with the registration button updated to reflect the student's current registration status once completed.

Figure 30 displays the interface of the student profile and attendance records interface for the Student module.

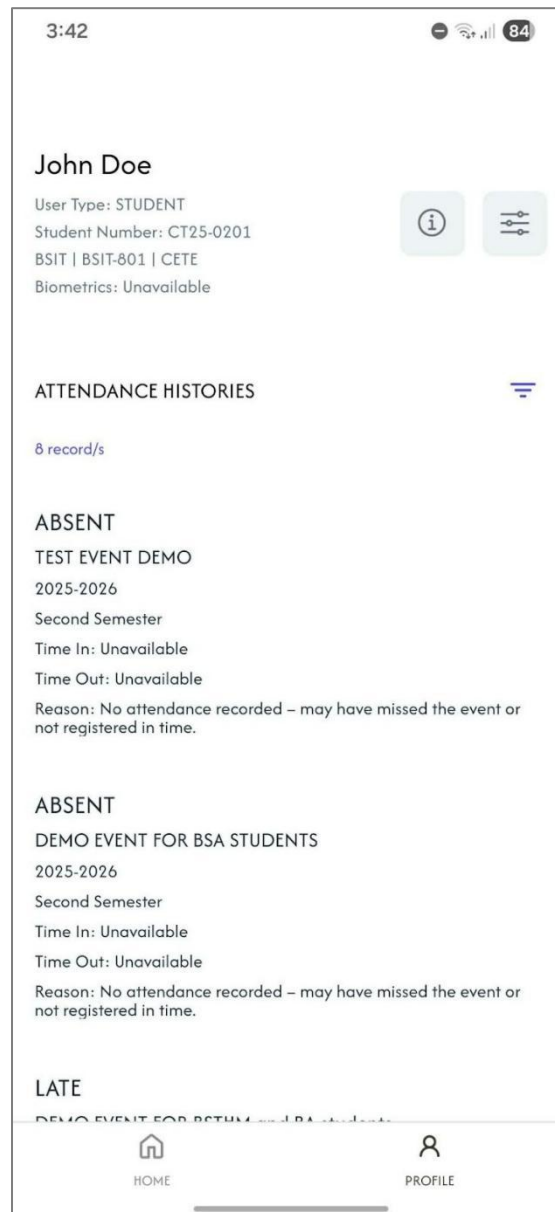


Figure 30. Student Profile and Attendance Records Page Interface

The figure displays the Student Profile and Attendance Records interface, which allowed students to view their personal account information and biometric status. It also provides a comprehensive attendance history feed showing their attendance status across all associated events in chronological order.

Figure 31 displays the settings page of the Student module.

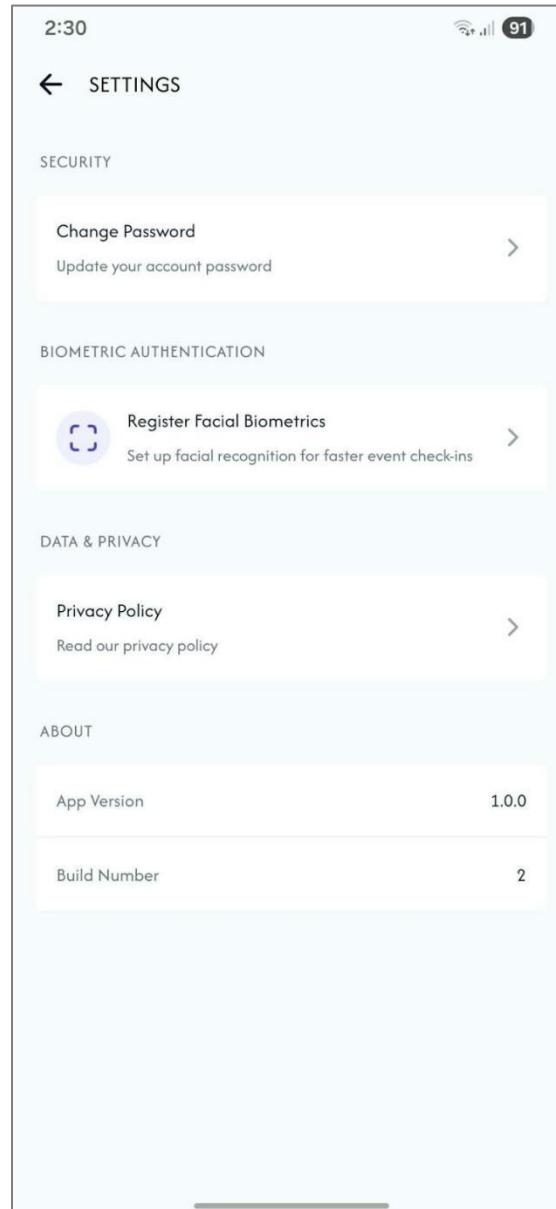


Figure 31. Settings Interface

The figure displays the Settings interface of the Student module, which allows students to manage their account security, biometric authentication, and privacy preferences, as well as view application information.

Project Evaluation

Following the development, the RCians Attendease underwent two (2) evaluations: a preliminary and final evaluation to ensure that the criteria set for each quality feature were met. Initially, the developers together with their adviser, conducted a preliminary evaluation in which several issues were identified. Issues such as, text colors for error, success, and information messages were not distinct, causing confusion for users. The automatic creation of course sections, which was not suitable for irregular students and larger sets of sections. And lastly, with the importing of student accounts error message, as the message it prompted wasn't understandable enough for the users, which the panel advised to replace it with more specific error messages like duplication, invalid password, etc. alongside the corresponding row number and student number of each affected student account. These observations highlight areas for improvement to make the system more reliable and user-friendly.

The developers used the ISO 9126 standard for the system assessment. For the final evaluation, the evaluators included are one (1) representative from the Office of Student Affairs for the web application, and 10% of the total population of enrolled college students for the mobile application.

Table 17 presents the quality factors for each criterion, along with their corresponding interpretations. The computation of the system's evaluation is presented in the appendix.

Table 17. Summary of End-User's Evaluation Results

Criteria	Quality Factor	Interpretation
Functionality	0.96	Excellent
Reliability	0.86	Excellent
Usability	0.93	Excellent
Efficiency	0.85	Excellent
Maintainability	0.94	Excellent
Portability	0.88	Excellent
Overall	0.92	Excellent

In the quality factor (QF) evaluation, the functionality of the system received a score of 0.96, interpreted as excellent. The sub-characteristics evaluated under functionality were suitability, accuracy, and security. This shows that the system performed its intended tasks correctly, met the user needs, and ensures that information is secure. However, slight differences in user ratings may be related to how the system performed in different usage situations. Some respondents may have experienced small changes in result accuracy location signal or internet connection were unstable. Others may have rated a system differently depending on how well its features matched their specific tasks or how clear they understood the security controls. Even with these differences, the system was still able to perform its main functions effectively.

The reliability criterion received a score of 0.86, also interpreted as excellent. This indicates that the system can operate consistently over time and under different conditions. Users can trust that tasks will be completed without errors and that the system will function as expected. However, some respondents may have observed minor inconsistencies and noted minor issues, such as occasional delays or the need to restart the application when navigating certain views. These small inconsistencies slightly affected the overall rating. Despite this,

the system remains highly dependable for regular use. The results indicate that the users can trust the system's performance.

For usability, the system scored 0.93, interpreted as excellent. Usability was assessed through understandability, learnability, operability, and attractiveness. This means the system is easy to understand, simple to learn, easy to operate, and provides a positive experience for its users. However, some users still asked about what to do next despite the proper demonstration beforehand, or they needed more time to get used to less familiar features. Minor issues like these may have slightly lowered the score. The system is mostly user-friendly, with just a few areas to refine.

The efficiency of the system scored 0.85, interpreted as excellent. Efficiency considered time behavior and resource utilization. The system responds quickly, uses resources effectively, and performs smoothly even when handling multiple tasks. However, some users experienced longer installation times, and the system's performance was affected by the stability of the internet connection. In addition, users with limited device storage sometimes had to remove other apps before installing the system, which slightly impacted efficiency. Minor delays like these may have lowered the score. Despite these issues, the system remains efficient for regular use.

The maintainability criterion received a score of 0.94, interpreted as excellent. This included changeability and testability. It shows that the system can be easily updated or improved, and any errors can be quickly detected and fixed when needed. However, implementing changes in some components may still

require careful testing to maintain overall system stability. Some features are slightly more complex to modify, certain tests may take extra time to complete. The system is highly maintainable, with just a few areas that could be refined.

The portability of the system scored 0.88, interpreted as excellent. Portability was assessed through adaptability, installability, and replaceability. This indicates that the system can run on different devices or platforms, can be easily installed, and can adapt to changes without difficulty. However, the mobile application is currently limited to the Android platform, as iOS development was beyond the scope of the study due to cost and licensing constraints. Despite this limitation, the system remains highly portable and adaptable to most user environments.

Overall, the system achieved a quality factor of 0.92, which is considered excellent. This means the system is reliable, easy to use, efficient, easy to maintain, and adaptable. It meets the standards for functionality, performance, and user satisfaction. Users can trust that the system will work as intended and support their tasks. The results also show some differences in user experience, such as device performance or internet connection. These small variations may have slightly affected some ratings. Even with this, the system is still high-quality and can be further improved for an even better user experience.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter includes a summary of findings, conclusions, and recommendations.

Summary of Findings

This study presents the development of RCians Attendease: An Automated Attendance Monitoring System for the College Department of Rogationist College, involving both the Office for Student Affairs (OSA) and the student body in its implementation. The system consists of two (2) interconnected modules: the OSA module, a web-based interface that enables the OSA to manage student accounts and attendance records, organize and manage events, configure locations, and oversee attendance data; and the Student module, a mobile application that allows students to monitor their own attendance. Development followed the Waterfall methodology of the Software Development Life Cycle (SDLC), covering all phases from requirements analysis, system design, development, testing, deployment, and maintenance, ensuring a clear and structured development process.

The RCians Attendease was evaluated by ten percent (10%) of the total college student population and one (1) representative from the OSA using the International Organization for Standardization (ISO) 9126 Quality Model to measure key quality attributes. The evaluation showed that the system scored 0.96 in functionality, 0.86 in reliability, 0.93 in usability, 0.85 in efficiency, 0.94 in

maintainability, and 0.88 in portability, achieving an overall quality factor of 0.92, classified as excellent.

These results indicate that the automated attendance monitoring system successfully met its intended objectives, delivering a reliable, user-friendly, and efficient solution for monitoring student attendance. The system effectively supports both students and the OSA in managing and maintaining accurate attendance records.

Conclusions

Based on the evaluation results and the findings gathered during the study, the following conclusions have been reached:

RCians Attendease: An Attendance Monitoring System using Geofencing Technologies was developed for the College Department of Rogationist College. A web-based module was created for the Office of the Student Affairs (OSA), and a mobile application was developed for students to record their attendance during events. Students can register their attendance using the RCians Attendease mobile application and view upcoming, registration, or ongoing events. The OSA can manage and monitor events, as well as view and print reports for documentation purposes. The use of geofencing technology proved effective in verifying student presence within designated event locations, providing a reliable and automated alternative to manual attendance tracking.

Through iterative testing and evaluation, the system was refined and validated as a functional tool for managing student attendance at institutional

events. To support end users, a user manual was provided containing step-by-step instructions to guide both the OSA coordinator and students in using the system effectively.

Recommendations

Considering the scope and limitations, as well as all the remarks, suggestions, and comments obtained during the system evaluation, the developer propose the following recommendations to further enhance the automation of attendance monitoring:

On system usability, deploying the mobile application to iOS is recommended to improve portability. The application's React Native framework makes iOS deployment technically feasible without requiring additional native code development. However, due to the high deployment costs and licensing fees associated with iOS development including Apple Developer Program membership, the current implementation is limited to the Android platform. Expanding both platforms would significantly broaden the system's reach, particularly given that it is intended for all college students at Rogationist College.

On connectivity and performance, mobile application users should maintain a stable internet connection, as core functionalities depend on real-time data synchronization with the server. Upgrading the network infrastructure and cloud service deployment is recommended to support higher traffic volumes and ensure system stability. Improving the existing backend

caching mechanisms would further reduce API response times, lower server load, minimize database read and write operations, and enhance overall application performance especially during peak usage periods.

On student engagement, implementing a notification system is recommended to improve communication between the system and its users. The mobile application should support push notifications to alert students when new events are created, when events are approaching or have ended, and when other important updates are available.

On system architecture and access control, module responsibilities should be separated into dedicated domains rather than consolidated under the Office of Student Affairs. The authorization and authentication system should be enhanced by clearly defining and refining user roles, leveraging an established identity and access management solution rather than reimplementing problems that have already been solved, to improve security, scalability, and maintainability. An administrator module should handle all user management tasks, including managing user accounts and information, along with academic management responsibilities such as managing academic years, clusters, courses, and sections. The OSA module should focus exclusively on event management and monitoring, attendance records management with the ability to update and delete records, and location management. Additionally, a student academic organization module should be introduced with limited access, allowing organizations to retrieve and request scheduled event creation pending OSA approval, create and retrieve locations, and view attendance records in read-only mode.

On backend architecture, the current implementation adopted a three-tier architecture, which provided an initial organizational structure suitable for the system's early development phase. However, as the system grew in complexity, maintaining clear boundaries between layers became increasingly challenging, making the codebase harder to extend and test independently. It is recommended that future development consider migrating toward a Clean Hexagonal Architecture, also known as Ports and Adapters, which enforces strict separation between core business logic and external dependencies. This architectural approach isolates the domain from infrastructure concerns, improving testability, maintainability, and resilience to change.

These improvements would collectively enhance the system's security, accountability, role-based access control, and overall operational efficiency by establishing clear boundaries between administrative, operational, and organizational functions.

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APPENDICES

APPENDIX A

EVALUATION INSTRUMENT

Instruction: Please evaluate the system by using the given scale and write the corresponding rating.

Numerical Scale	Interpretation
3	Excellent
2	Good
1	Fair
0	Poor

Characteristics	Sub – Characteristics	Description	Rating
Functionality	Suitability	The application meets user needs and requirements.	
	Accuracy	The application provides correct and precise results.	
	Security	The application protects data and controls access.	
Reliability		Application is able to perform consistently and dependably.	
Usability	Understandability	Users can easily comprehend the application's functions.	
	Learnability	Users can quickly learn to use the application.	
	Operability	Users can effectively operate the application.	
	Attractiveness	The visual of the application is appealing to end users.	
Efficiency	Time Behavior	The application has short response time and speed of performance.	
	Resource Utilization	The application uses available resources.	
Maintainability	Changeability	The application can be easily change and update without affecting other parts of the application.	

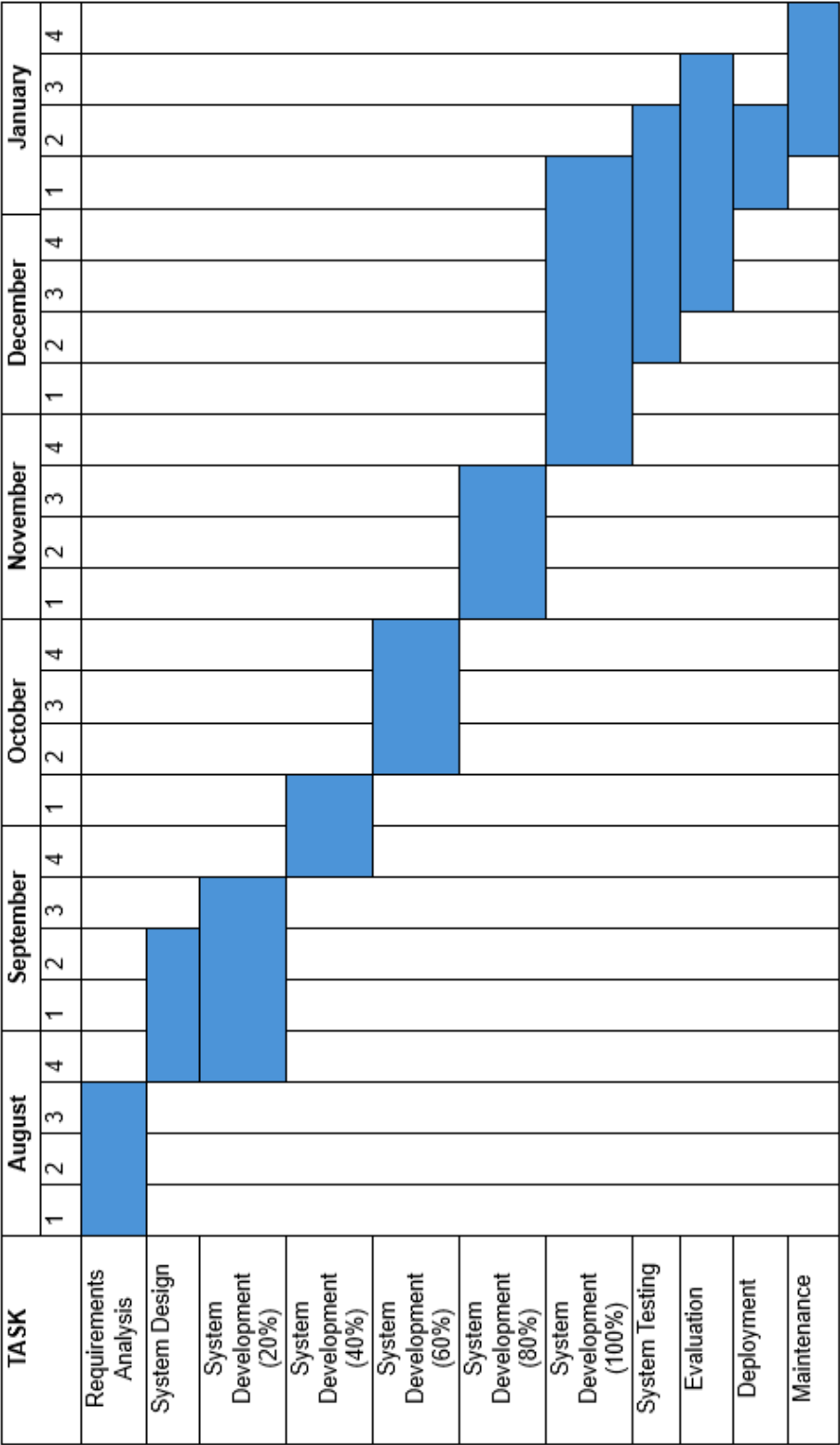
	Testability	The application can be easily tested for defects and compliance.	
Portability	Adaptability	Application is able to function in different environments.	
	Installability	The application can be easily installed by the user.	
	Replaceability	The application can be easily replaced with another.	

Remarks:

Evaluated by: _____
 Signature over printed name

APPENDIX B

GANTT CHART



APPENDIX C

CONSENT LETTER FOR EVALUATION FOR STUDENTS

Participants Consent Form: RCIANS ATTENDEASE

The Lord be with you and good day!

We, the Researchers from BSIT-801, are currently conducting a Capstone Project entitled **“THE DEVELOPMENT OF RCIANS ATTENDEASE: AN ATTENDANCE MONITORING SYSTEM USING GEOFENCING AND FACIAL RECOGNITION TECHNOLOGIES FOR ROGATIONIST COLLEGE - COLLEGE DEPARTMENT.”** As a vital part of our research, we are developing a system that includes a mobile application designed to ease the students’ attendance registration and tracking their records.

Hence, we are inviting you to participate in our research study involving the testing of a mobile application designed to streamline event attendance registrations and attendance records tracking. Before you proceed, please read the following information regarding the data we collect and how it is used.

Required Permissions & Data Collection To participate in this testing phase, the application will require access to:

- **Camera Access:** For facial registration and identity verification. (Note: Your facial data is used solely for identity matching within the system).
- **Location Services:** To ensure you are within the designated "Geofenced" area (event registration and event venue) when marking attendance.
- **Internet Connection:** Required to be used for real-time data synchronization.

Confidentiality, Data Privacy and User Control:

- All data collected (name, student number, facial encodings, and location logs) will be treated with the strictest confidentiality.
- Data will be stored on a secure server and will only be accessible to the research team.
- The data will be used solely for academic research and will be deleted/destroyed upon the completion and approval of the thesis.
- You maintain full control over your biometric information in accordance with Republic Act No. 10173 (Data Privacy Act of 2012). You may freely remove your registered facial data at any time directly through the application settings.

- Voluntary Participation Your participation is entirely voluntary. You may withdraw your consent and stop using the application at any time.

Participant Agreement By signing below, I acknowledge that:

- I have read and understood all the information above.
- I voluntarily agree to install the application and provide the necessary permissions (Camera and Location).
- I understand that my data will be used for research purposes only.

Participant Name (Printed): _____

Course & Year: _____ Date: _____

Signature: _____

APPENDIX D

COST BENEFIT ANALYSIS

**Cost Benefit Analysis for The Development of RCians Attendease:
An Attendance Monitoring System Using Geofencing and
Facial Recognition for the College Department of
Rogationist College**

Cost				
Category	Item	Price	Quantity	Total
Deployment	Cloud Service Deployment for Backend Services (Render)	PHP 644.76	1	PHP 644.76
	Cloud Service for Facial Recognition Service (Hostinger VPS)	PHP 509	1	PHP 509
	Cloud Service Production Deployment (AWS)	FREE	1	PHP 00
	Domain	PHP 70.23	1	PHP 70.23
Other	Bond Paper	PHP 238	2	PHP 476
	Bond Paper	PHP 358	1	PHP 358
Total Cost				PHP 2,057.99

APPENDIX E

COMPUTATION OF QUALITY FACTOR

CHARACTERISTICS	SUBCHARACTERISTICS	RATING																												QVC	FQVC	TQVS	ITQVS	QF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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CHARACTERISTICS	Description
Functionality	Excellent
Reliability	Excellent
Usability	Excellent
Efficiency	Excellent
Maintainability	Excellent
Portability	Excellent
Overall:	0.92 - Excellent